

CSG Racing Electrical Assessment

Sections A, B, C and D

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Interests: LV hardware > Embedded > HV

Summary

I completed Sections A, B, C and D. In A, I proposed the vehicle electrical layout, reviewed the shutdown circuit and calculated precharge behaviour. In B, I checked resistor ratings. I also simulated normal charging, an open path and a shorted DC link. In C, I designed a coil driver and a protected 5 V feedback input. I tested six operating and fault cases in simulation. In D, I wrote the shutdown logic and ran the supplied native tests. I then checked the outputs and door/tamper button in Wokwi. The diagrams and results appear beside the relevant answers. The appendices contain the code, component list and test instructions. This work does not include physical hardware tests.

I use the FSUK 2026 v1 rulebook specified in the assessment. I state design assumptions and any hardware checks still needed. References [1]–[16] are external sources. My diagrams, simulations and code are listed separately as project evidence.

Sections: A.1 architecture; A.2 design review; A.3 precharge; A.4 source review; B.1 resistor rating and fault detection; B.2 accumulator safety; C.1 driver; C.2 feedback protection; D.1 embedded task.

Section A Core assessment

A.1 High level system overview

Design basis and assumptions

The assessment specifies 32 LFP blade cells in series. Pack voltage is 102.4 V nominal and 116.8 V maximum. It uses an Orion BMS 2 and a nominal 12 V low-voltage supply. The regulated rails are 5 V and 9 V. CAN connects the BMS, inverter, dashboard and logger. A service disconnect and pre-charge path are available. [1, p.2]

I propose the layout below for a manually driven electric vehicle. I do not assume an autonomous system. I use two normally open mechanical AIRs, one in each battery pole. The pre-charge relay is also normally open. These choices follow the rules. I do not copy the relay choice in the historical paper. [2, EV5.6.1–EV5.6.3, EV5.7.2]

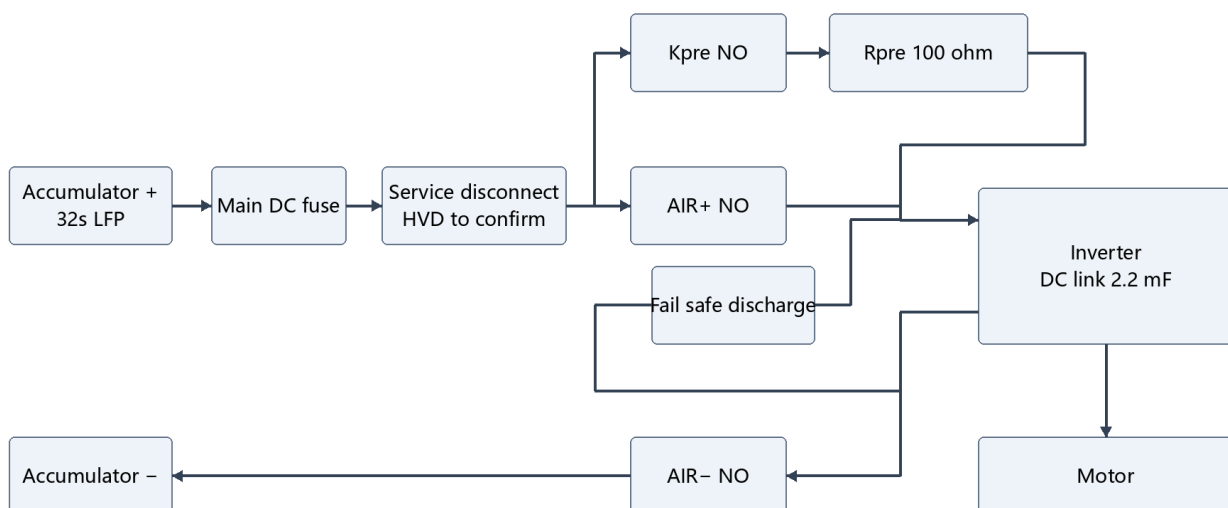
A fused LV battery powers the safety and control electronics before TS activation. An isolated DC-DC converter can support the LV supply when its TS input is available. The team must confirm its connection point, start-up behaviour and isolation. I do not rely on this converter alone to start pre-charge. [Design assumption; 2, EV1.1.1–EV1.2.1]

I measure pack and DC-link voltage through isolated or suitably rated interfaces. Contact feedback checks the actual contactor state. The inverter stays disabled until pre-charge and the ready-to-drive checks are complete. Cell limits, BMS outputs and fuse ratings need confirmation. The contactor models, measurement accuracy and timeouts also need checking. [Design proposal; 2, EV4.11, EV5.7.1, T11.9]

Power path

This is a functional diagram, not a wiring or manufacturing schematic. The service-disconnect block is a concept. The team must confirm that it meets the HVD requirement and check its physical layout. TS negative is not connected directly to LV ground. [1, p.2; 2, EV1.1–1.2, EV4.8]

Figure 1 shows the proposed power path.



Functional concept TS negative is not LV ground Discharge spans both DC link poles

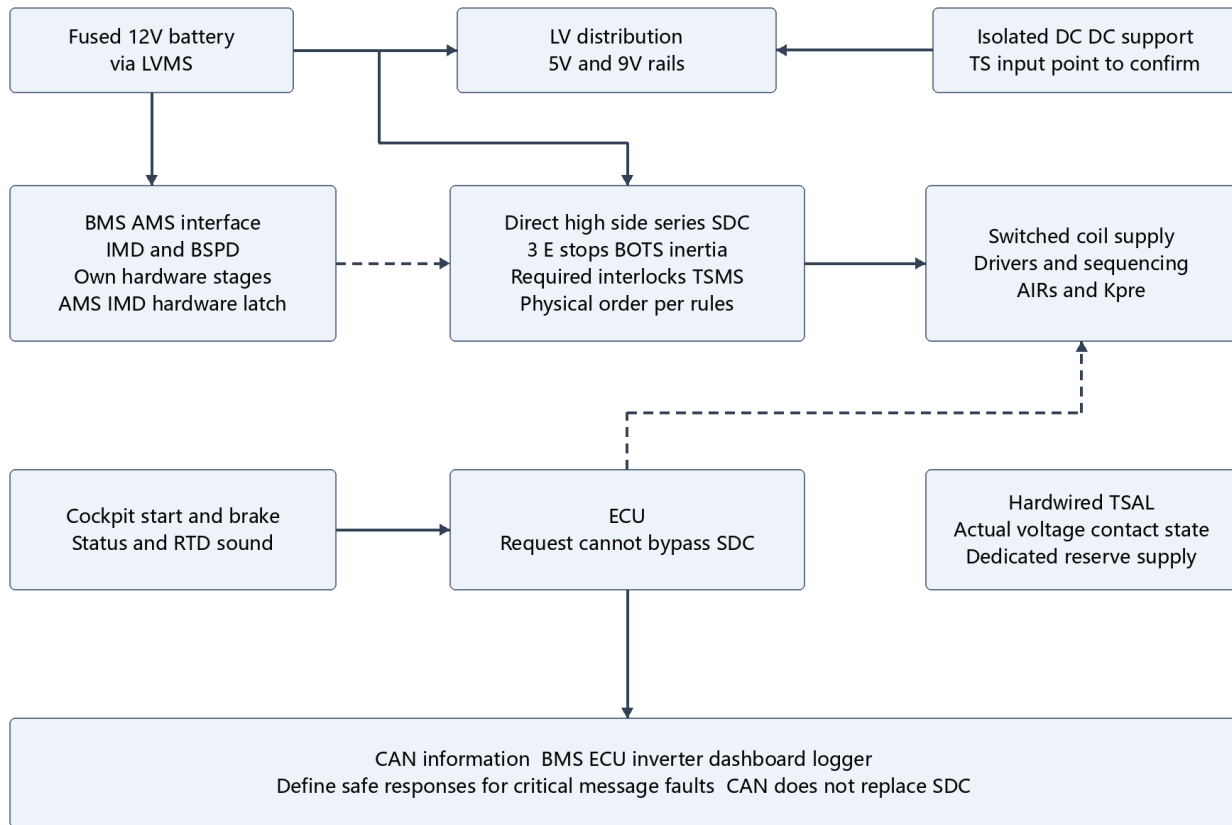
Figure 1. Proposed tractive-system power path.

Note: My architecture drawing. AIRs and Kpre are normally open. The team must confirm the service-disconnect/HVD arrangement. TS negative is not LV ground.

The accumulator feeds the inverter through a fuse and two-pole isolation. The resistor and pre-charge relay bypass AIR+. AIR– provides the return path. AIR+ closes only after the measured vehicle-side voltage meets the pre-charge condition. Kpre then opens. I would check the exact sequence and switching overlap with the selected contacts.

LV power shutdown and information paths

Figure 2 shows the LV supply, shutdown and communication paths.



Safety power and information paths shown separately Functional concept only

Figure 2. Proposed LV supply, hardware shutdown and CAN information paths.

Note: My architecture drawing. Dashed links show requests or hardware interfaces. CAN does not replace direct SDC interruption. The diagram does not set the physical order of the switches.

The dashed safety links show each device acting through its own hardware stage in the SDC. They are not CAN commands. The SDC directly carries the AIR and pre-charge drive supply on the high side. Shutdown buttons, BOTS, TSMS and interlocks act directly. They do not use a firmware-controlled output stage. The TSMS must be placed as EV6.1.4 requires. This block diagram does not set the full physical order. [2, EV6.1.1–EV6.1.4, 1.9–1.10]

The ECU can request a contactor sequence. However, its request must not keep a coil powered after the SDC opens. Other supplies and interfaces must not backfeed the switched rail. AMS and IMD trips use non-programmable latches. The required manual reset is outside the driver's control. [2, EV6.1.6–EV6.1.9]

TSAL has its own voltage and contact detection. It also needs the required reserve supply. I show it as a separate safety function, rather than an ECU-controlled dashboard icon. [2, EV4.10.5–EV4.10.6, 1.9, 1.16]

Purpose classification failure and safe response

An interface block crosses the TS and LV domains. Its pins may therefore have different voltage classifications. The responses below are design targets for testing. Opening a coil driver does not resolve every possible fault.

Block	Purpose and domain	One credible failure	Expected safe response and basis
Accumulator/cells	Store energy; TS	Cell overvoltage or excessive temperature	AMS opens SDC and latches the trip; inhibit restart until diagnosed. Cell limits come from the cell datasheet and rules. [2 EV5.8, EV6.1.6]
Main fuse	Protect the TS current path; TS	Short circuit causing excessive current	Properly coordinated DC fuse interrupts within its specified capability; SDC alone is not the short-circuit protection. [2 EV3.2]
AIR+ and AIR–	Disconnect both battery poles; TS contacts/LV coils	One main contact welds closed	Remove drive and request all isolation devices open. Inhibit restart and check voltage/contact mismatch. Do not assume safe isolation. Investigate with the other pole open. [2 EV5.6, EV4.10.5; proposed diagnosis]
Pre-charge branch	Limit capacitor charging current; TS power/LV control	Resistor or relay path fails open	Measured DC-link voltage fails to rise; abort on a verified timeout, leave the second AIR open and remove coil requests. [2 EV5.7; proposed timeout]
Inverter	Convert DC to motor drive; TS/LV interface	Internal power-stage fault	Remove torque/enable and activate the relevant protective response; use rated fault-current protection. Do not rely on a CAN instruction to interrupt a hard short. [2 EV3.2; design response]
Motor	Convert electrical energy to torque; TS	Insulation fault to chassis	IMD opens SDC, inhibit torque and latch the fault; physical motion is handled by the mechanical brakes. [2 EV6.3, EV6.1.6]
Orion BMS 2 / AMS	Monitor cells, temperature and current; interface	Supply or measurement connection lost	Loss of permission opens SDC. Measurement loss must trip. Check supply-loss behaviour and the chosen output configuration. [2 EV5.8.10, EV6.1.7; 6]
IMD	Detect TS insulation faults; interface	Insulation or IMD failure	Open SDC without programmable influence and latch the trip. [2 EV6.3.7, EV6.1.6]
BSPD	Detect specified hard-braking/power implausibility; LV	Sensor signal disconnected	Detect its SCS failure and open SDC with standalone non-programmable circuitry. [2 T11.6, T11.9]
Shutdown circuit	Directly remove coil supply; LV safety path	Wire breaks	De-energise AIR and pre-charge drivers; a separate short-circuit bypass scenario must also be reviewed. [2 EV6.1.1, 1.7; T11.9]

Block	Purpose and domain	One credible failure	Expected safe response and basis
LV battery/distribution	Supply safety/control equipment; LV	LV voltage collapses	Inhibit drive and default coil supply to off; prevent unsafe automatic restart on recovery. TSAL reserve operation is handled separately. [2 EV6.1.7, EV4.11.4, EV4.10.16]
Isolated DC-DC	Support LV supply from TS; interface	Output fails low	Use remaining valid LV supply only if safe conditions remain satisfied; otherwise shut down. Confirm isolation and protection for other failures. [2 EV1.1–1.2; design assumption]
5 V/9 V regulators	Supply logic/sensors; LV	Regulator undervoltage resets ECU	Inhibit torque and restart; protected safety interfaces must not interpret invalid outputs as permission. [2 T11.9; design response]
Three shutdown buttons	Allow manual emergency opening; LV/direct SDC	Contact/wiring fails open	Keep SDC open and prevent activation. Test each button. A stuck-closed contact needs a separate fault review. [2 T11.4, EV6.1.10]
BOTS and inertia switch	Detect overtravel/impact; LV/direct SDC	Actuation opens contact	Open SDC; inertia function latches until reset. [2 T6.2, T11.5, EV6.1.2]
HVD/required interlocks	Permit physical separation and detect disconnection; TS with LV interlock	HVD removed	Interlock opens SDC; discharge path still operates with the battery disconnected. Pack interiors remain energised. [2 EV4.8–4.9]
ECU/cockpit interface	Process driver requests and operating state; LV	ECU freezes with an old torque request	Inverter timeout/enable controls inhibit torque; independent SDC remains effective. Opening SDC immediately exits ready-to-drive. [2 T11.9, EV4.11.8; proposed implementation]
CAN nodes	Exchange status and commands; LV communication	Safety-critical message lost/delayed	After a justified timeout, treat critical data as invalid and enter the defined safe state. Check dependencies before making logger loss a shutdown condition. [2 T11.9.1–1.5]
DC-link voltage/contact feedback	Verify pre-charge and actual state; interface	Sensor gives an implausible high value	Reject measurement, prohibit second-AIR closure and diagnose; test stuck readings as well as disconnection. [2 EV5.7.1, T11.9; proposed diagnostic]
Discharge circuit	Reduce residual capacitor voltage; TS	Discharge path open	Voltage monitoring detects decay failure; keep system disabled and indicate that deactivation is unconfirmed. [2 EV4.9, EV6.1.5; proposed diagnosis]

Block	Purpose and domain	One credible failure	Expected safe response and basis
TSAL and cockpit indicators	Communicate TS/fault state; interface	Contact-state detection implausible	Follow the indicator fault requirements. Extinguish green where required. A dark lamp does not prove zero voltage. [2 EV4.10.11–EV4.4.10.15, T11.9.5]

Operating sequence and verification before release

1. Power the LV electronics. Keep AIRs and Kpre open and disable inverter torque. Check the safety inputs and measurements.
2. Obtain a deliberate TS activation request. Closing the SDC alone must not activate the TS. [2 EV4.11.4]
3. With safety permission present, close AIR– and Kpre. Charge the DC link through Rpre. Check this proposed sequence against the actual hardware.
4. Compare measured DC-link and pack voltages. Allow AIR+ to close only at a guaranteed threshold of at least 95% of actual pack voltage. Include sensor error. Check the transition, then open Kpre. [2 EV5.7]
5. Enter ready-to-drive only after the required extra driver action while braking. Produce the specified RTD indication. [2 EV4.11.7, EV4.12]
6. When the SDC opens, exit ready-to-drive and remove AIR/Kpre supply through the hardware path. Check the required voltage decay. Any AIR delay must meet EV6.1.5. Do not copy the delay from a historical waveform. [2 EV4.11.8, EV6.1.5]

Before release, I would confirm the cell limits and BMS outputs. I would check the contactors' DC ratings and opening behaviour, resistor pulse rating, and inverter capacitance and load. I would also check DC-DC isolation, fuse coordination and voltage-sensing accuracy. TSAL reserve supply, wiring and interlocks need review. Tests would use simulation and safe LV fault injection. They would cover open/short signals, MCU freeze/reset, supply loss and lost CAN messages. I would also test pre-charge timeout and contact feedback mismatch. This written assessment does not require accumulator or HV testing. [1 p.1; 2 T11.9, EV6.1.8; proposed verification plan]

A.2 Design review

I would not use a microcontroller as the only path from the safety switches to AIR energisation. A firmware hang or brownout could block shutdown. So could an incorrect input reading or an output stuck on. A watchdog helps with some processor faults. It cannot fix a shorted output stage, a bypassed switch or backfeeding from another supply. [Design fault analysis; 2 T11.9]

FSUK v1 requires the SDC to carry AIR/pre-charge drive power directly. The specified chain must be on the high side. Shutdown buttons, BOTS, TSMS and interlocks must act directly, without a power stage. Each system that can open the SDC needs its own non-programmable stage. AMS and IMD trips need non-programmable latching. BSPD must be standalone and non-programmable. IMD shutdown must not depend on programmable logic. [2 EV6.1.1–EV6.1.3, 1.6, 1.9–1.10; T11.6.1; EV6.3.7]

I would use software for diagnostics, logging, displays and communication checks. It can also coordinate start-up when the hardware permits it. It must not bypass hardware shutdown or restart TS automatically when the safety chain recovers. A programmable BMS can still be used. Its independent output stage and latch must meet the rules, and the interface must be demonstrated. [2 EV5.8, EV6.1.6–EV6.1.9, EV4.11.4; design proposal]

Before accepting the design, I would map each rule to the relevant circuit. I would define safe states and review SCS faults. I would check output specifications, boot/reset behaviour and power loss. Response-time budgets and fault-injection results are also needed. Each shutdown request must still work with the MCU frozen. Tests must show that another supply cannot keep the coil rail powered. I would therefore retain the compliant hardware shutdown chain and use software within that boundary. [2 T11.9.7, EV6.1.8–EV6.1.9; proposed acceptance evidence]

A.3 Contactor and pre-charge design

Inputs and model

From the assessment: $C=2.2\text{ mF}=0.0022\text{F}$, $V=116.8\text{ V}$, $V_c(0)=0\text{ V}$ and $R=100\ \Omega$. [1, p.3]

I assume a constant-voltage source and fixed resistance and capacitance. Extra series impedance is negligible. The inverter draws no load during pre-charge, and there is no leakage. The equations are:

$$V_c(t) = V \times [1 - \exp(-t / RC)]; I(t) = (V / R) \times \exp(-t / RC)$$

I use an ideal first-order RC charging model. Its purpose matches the pre-charge discussions in [3], §2.5.1 and [4], §2.4.

1 Initial current

$$I_0 = [V - V_c(0)] / R = 116.8 / 100 = 1.168\text{ A}$$

2 Initial resistor power

$$P_0 = I_0^2 R = 116.8^2 / 100 = 136.4224\text{ W} \approx 136.4\text{ W}$$

This is the initial instantaneous power. During ideal charging, $P(t) = P_0 \exp(-2t / RC)$. The initial value does not directly set the resistor's continuous rating.

3 Time to 95 percent

$$\tau = RC = 100 \times 0.0022 = 0.22\text{ s}$$

$$0.95 = 1 - \exp(-t / 0.22); t = -0.22 \ln(0.05) = 0.659061\text{ s} \approx 0.659\text{ s}$$

At 116.8 V, the 95% threshold is 110.96 V. The real controller must use actual measured pack voltage. It must not use a fixed 110.96 V threshold at every pack voltage. The rule requires an intermediate-circuit voltage measurement before the second AIR closes. [2 EV5.7.1]

4 Two reasons not to close the main contactors immediately

Contact damage: An uncharged capacitor has a large voltage difference across the low-impedance main path. The resulting inrush can cause arcing or contact damage. It can also weld the contacts and defeat isolation. [4 §2.4; 3 §2.5.1]

Stress on the power path: The inrush also stresses the DC-link capacitors, battery, connections and conductors. Pre-charge limits this current. The given data cannot determine the unprotected peak because the bypass-path impedance is unknown.

Engineering interpretation

The 0.659 s result is an ideal RC charging time. I use measured bus voltage to decide when to enable the main path. A timeout detects failed precharge. Ideal resistor energy is $\frac{1}{2}CV^2 = 15.006464\text{ J}$. I use this in B.1 to assess the pulse rating. [1, A.3; RC model]

A.4 Evidence based design review

Selected source: Badawy et al., Figure 11, “AIRs and pre-charge relay circuit,” interpreted together with §2.5.1; PDF p.10, journal p.199. [3]

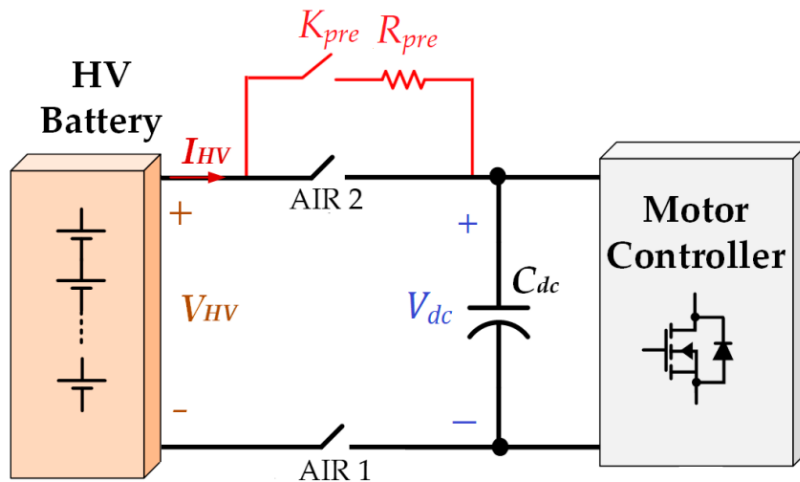


Figure 11. AIRs and pre-charge relay circuit.

Figure 3. Published AIR and precharge circuit from Badawy et al. [3], Figure 11, PDF page 10 / journal page 199.

Note: Reproduced under CC BY 4.0. This is an external source figure, not my circuit or a verified CSG design.

Figure 3 limits the inrush into an uncharged inverter DC-link capacitor. Without precharge, connecting it directly to the accumulator would produce a large current. A resistor and relay first provide a controlled charging path. The main positive contactor then closes. This protects the contacts and power path while allowing normal operation after charging. [3, Figure 11 and Section 2.5.1]

The design needs reliable measurements of DC-link and accumulator voltage. It also needs enough resistor pulse capacity, suitable contactor ratings and a clear switching sequence. The paper uses a much higher vehicle voltage than the CSG brief. It also describes a normally closed pre-charge relay. For my RC estimate, I assume negligible inverter current during charging. This is my modelling assumption, not a result demonstrated by the paper. It must be checked against the chosen inverter. [3, abstract and Section 2.5.1]

I would use the same basic idea: a current-limited branch around one main contactor. Voltage measurements then decide when to connect the full power path. With CSG's 116.8 V, 100-ohm resistor and 2.2 mF capacitor, initial current is 1.168 A. Initial resistor power is 136.4 W. The ideal time to 95 percent is about 0.659 seconds. These calculations guide sizing and tests. They do not form a complete hardware specification. [1, A.3; calculations in A.3]

Some parts need changing. FSUK 2026 v1 requires a normally open mechanical pre-charge relay. It also requires measured voltage of at least 95 percent of actual accumulator voltage before the second AIR closes. I would replace the paper's normally closed relay and check the loss-of-power response. CSG needs suitable voltage-sensing isolation, contactor feedback and resistor ratings. Shutdown timing also needs review. The paper reports about 280 ms of delay. This exceeds v1's permitted 250 ms AIR-opening delay. [2, EV5.7 and EV6.1.5; 3, Section 5.3]

Before using the circuit, I would calculate charging time with component tolerances and check resistor energy. I would check voltage-measurement error near the threshold. Simulation would cover an open branch, a shorted inverter input and an invalid voltage reading. Safe LV mock-ups could check relay sequencing, timeout and supply loss. I would then review shutdown and restart against the rules. The team must confirm the inverter and contactors. This assessment needs no HV construction or energisation. [1, assessment restrictions; 2, EV6.1 and T11.9]

Section B HV and accumulator

B.1 Electrical calculations

The assessment gives $V_{\text{pack}} = 116.8 \text{ V}$ maximum, $R = 100 \text{ } \Omega$, $C = 2.2 \text{ mF}$ and $V_{\text{dc}}(0) = 0 \text{ V}$. I assume a constant source and ideal resistor and capacitor. Normal inverter load is negligible.

Quantity	Calculation	Result
Initial current	$I_0 = V_{\text{pack}} / R$	1.168 A
Initial power	$P_0 = V_{\text{pack}}^2 / R$	136.42 W
Time constant	$\tau = RC$	0.220 s
Time to 95%	$t_{95} = -RC \ln(0.05)$	0.659 s
95% bus voltage	$0.95 \times V_{\text{pack}}$	110.96 V
Full-charge resistor energy	$E_r = \frac{1}{2}CV_{\text{pack}}^2$	15.01 J
Resistor energy to 95%	$E_r(t_{95}) = \frac{1}{2}CV_{\text{pack}}^2(1 - 0.05^2)$	14.97 J

$V_{\text{dc}}(t) = V_{\text{pack}}(1 - \exp(-t/RC))$; $I_{\text{pre}}(t) = (V_{\text{pack}}/R)\exp(-t/RC)$; $P_{\text{pre}}(t) = (V_{\text{pack}}^2/R)\exp(-2t/RC)$. FSUK 2026 v1 EV5.7.1 requires the measured DC-link voltage to reach at least 95% of the actual accumulator voltage before the second AIR closes [2].

B.1 Resistor rating recommendation

I assume at least 10 s between starts. Average full-charge dissipation is $15.01/10 = 1.50 \text{ W}$. A $2\times$ margin gives 3.00 W . I specify at least 5 W of usable continuous capacity after mounting and temperature derating. This needs a retry limit and cool-down control.

For each pulse, I use an energy target of $\geq 30 \text{ J}$ and a $2\times$ power-profile margin. $P_{\text{design}}(t) = 272.84 \exp(-2t/0.22) \text{ W}$. The resistor's allowed pulse envelope must cover the full event and repeat rate. Checking only the starting peak is not enough. I would also check working voltage, tolerance, heat sinking and ambient temperature. These margins are design assumptions, not rule limits.

Vishay LTO 100 H is a candidate family. Its overload graph does not confirm suitability for the full 0.66 s event [16]. Before choosing a part, I would check a matching manufacturer curve or equivalent rating data. I would ask the manufacturer if these data are insufficient. Sustained short-circuit heating needs a separate check against the protection cutoff time.

B.1 Multisim results and fault decisions

I ran three V2 netlists in Multisim: normal charging, an open precharge path and a shorted DC link. Node 3 is an ideal current-monitor output at 1 V/A . It does not load the circuit. Screenshot readings are approximate. The table's reference values are calculations, not cursor measurements.

Case	Observed behaviour	Calculated reference
Normal	Bus rises; current decays.	$t_{95} = 0.659 \text{ s}$; $V_{\text{dc}} = 110.96 \text{ V}$; $I_{\text{pre}} = 0.0584 \text{ A}$
Open	Bus near zero; monitor about 117 pV .	$R_{\text{pre}} = 10^{12} \text{ } \Omega$; $I_{\text{pre}} \approx 116.8 \text{ pA}$
Short	Bus settles near 1.156 V ; current persists near 1.156 A .	$R_{\text{fault}} = 1 \text{ } \Omega$; $V_{\text{dc}} = 116.8/101 = 1.1564 \text{ V}$; $I_{\text{pre}} = 1.1564 \text{ A}$

I represent the open path with a very large resistance. The short model places an assumed $1 \text{ } \Omega$ load across the capacitor. This is not a measured inverter impedance. Its time constant is $(100 \text{ } \Omega \parallel 1 \text{ } \Omega) \times 2.2 \text{ mF} = 2.178 \text{ ms}$. Current falls only slightly from 1.168 A . It does not approach zero.

The plots show normal charging and both fault cases. Node 1 is pack voltage. Node 2 is DC-link voltage. Node 3 is a current monitor at 1 V/A . The horizontal axis is time. Here, m means milliseconds and p means 10^{-12} .

Design and Multisim implementation

Figure 1 shows the intended precharge path. Figure 4 shows the open-path test in Multisim. It uses a 116.8 V source and a 2.2 mF capacitor. Rpre is set to 1 T Ω to model an open path. The normal calculation and netlist use 100 Ω . Rleak models leakage. Emonitor is a simulation measurement element, not a purchased part. The bus stays almost uncharged in this fault test, as the plots below show.

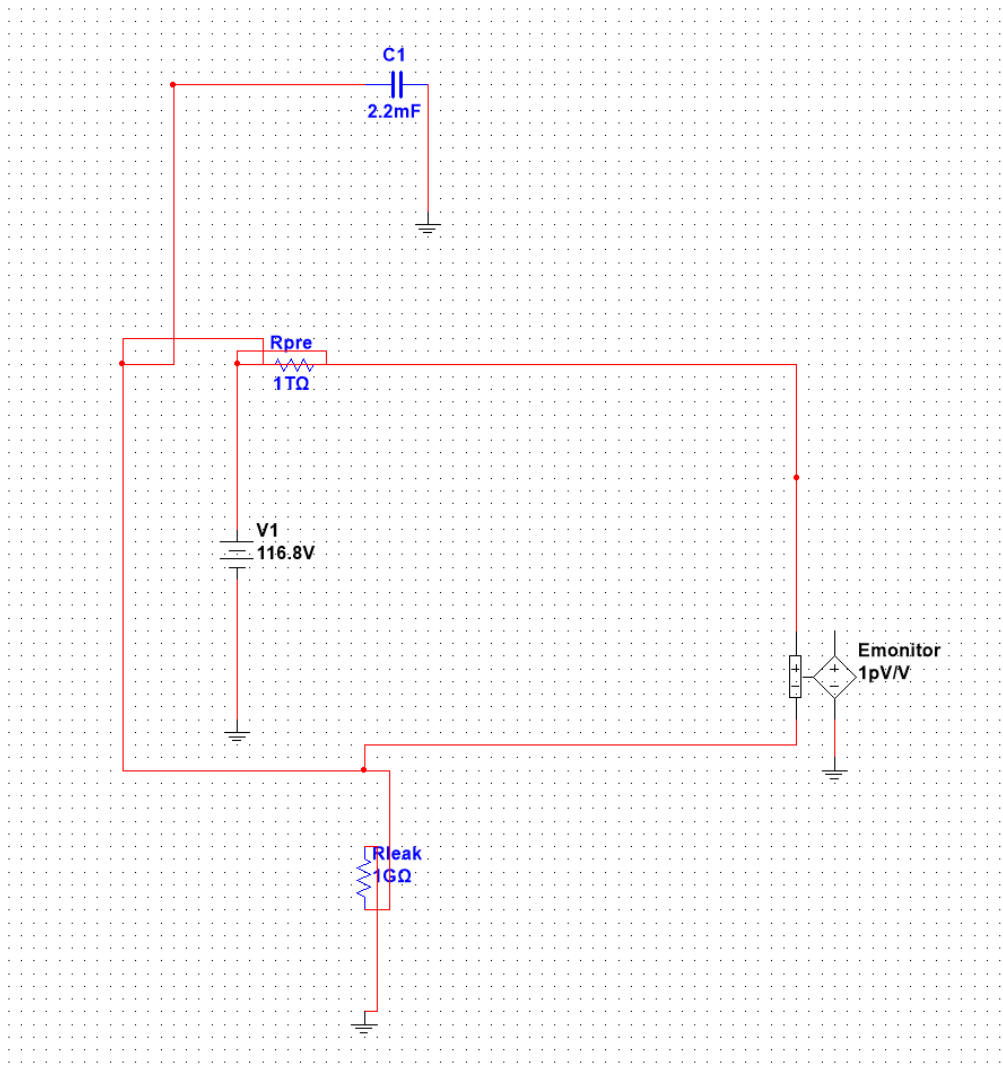


Figure 4. Multisim schematic view for the B precharge open-path fault test.

Source: applicant-supplied Multisim screenshot. This is a simulation schematic, not an energised physical circuit.

Normal precharge

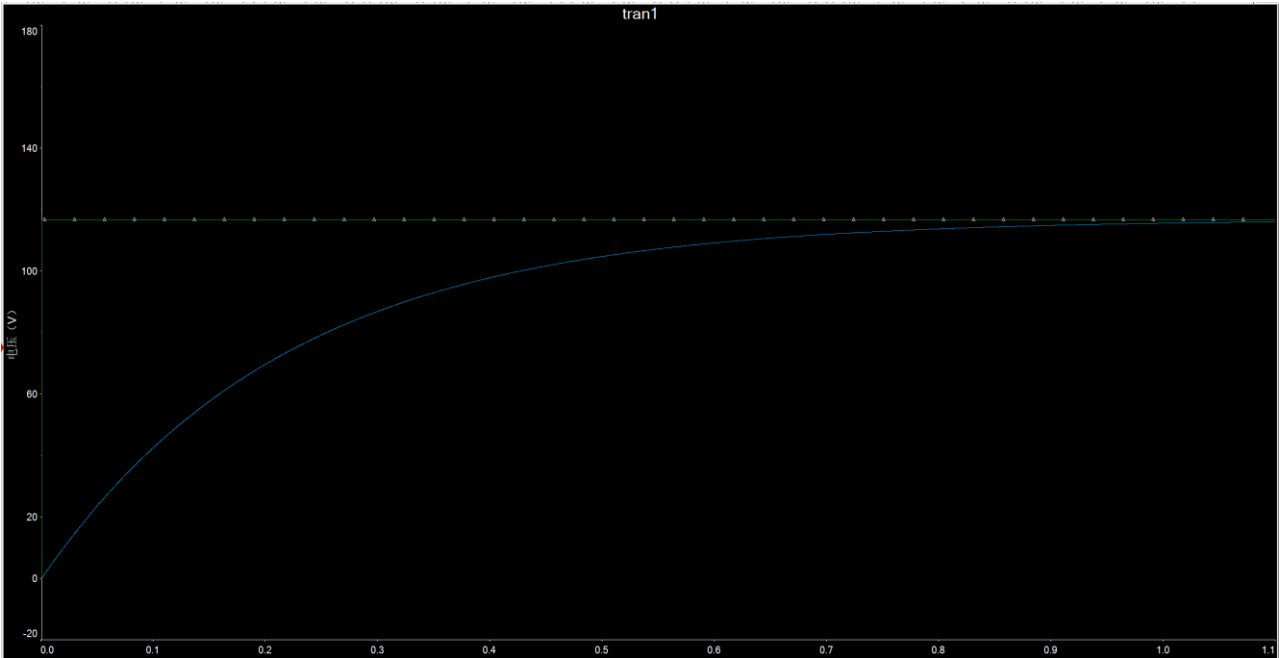


Figure 5. Pack remains near 116.8 V while the DC link charges.
Note: Multisim simulation. V(3) is a current monitor: 1 V represents 1 A.

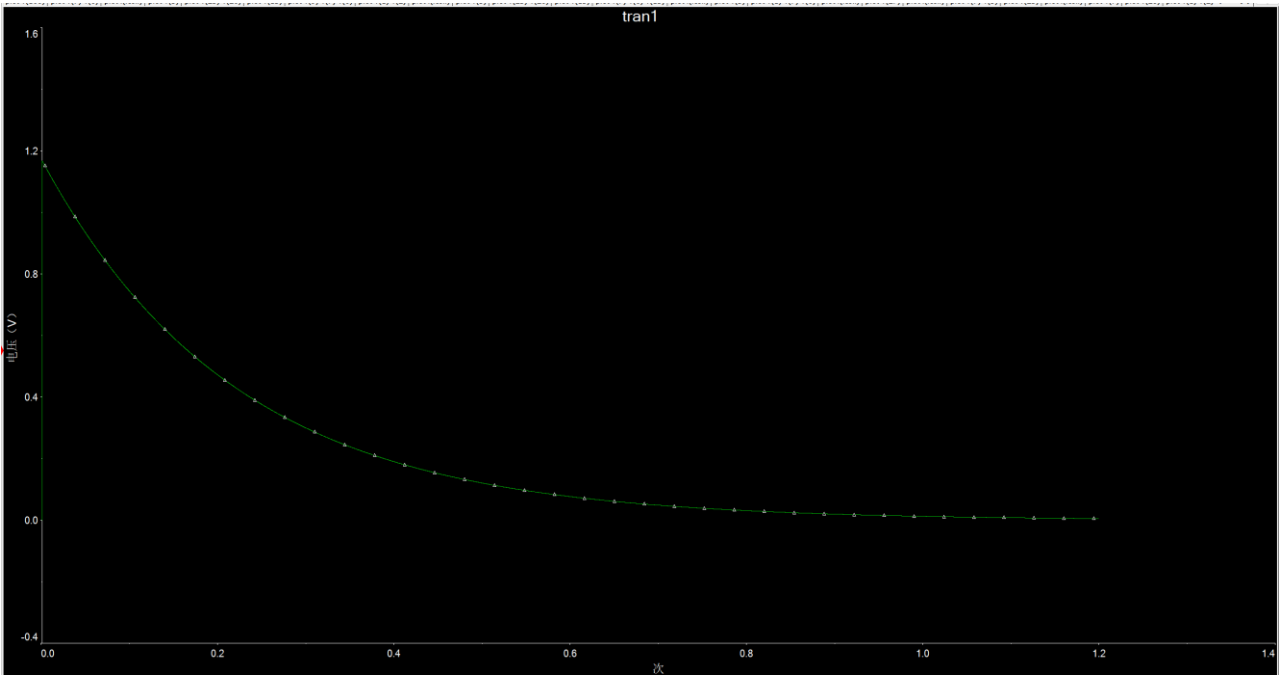


Figure 6. Current monitor decreases from approximately 1.168 V toward zero.
Note: Multisim simulation. V(3) is a current monitor: 1 V represents 1 A.

Open precharge path

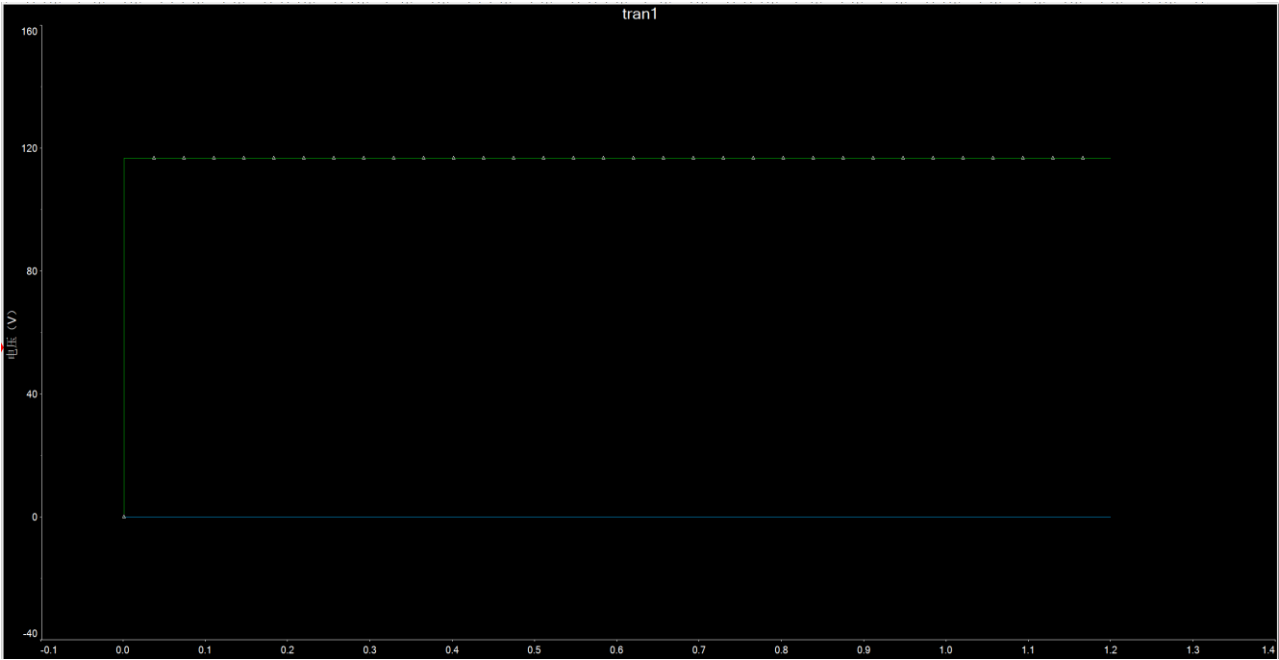


Figure 7. Pack voltage is present but DC-link voltage remains near zero.
Note: Multisim simulation. V(3) is a current monitor: 1 V represents 1 A.

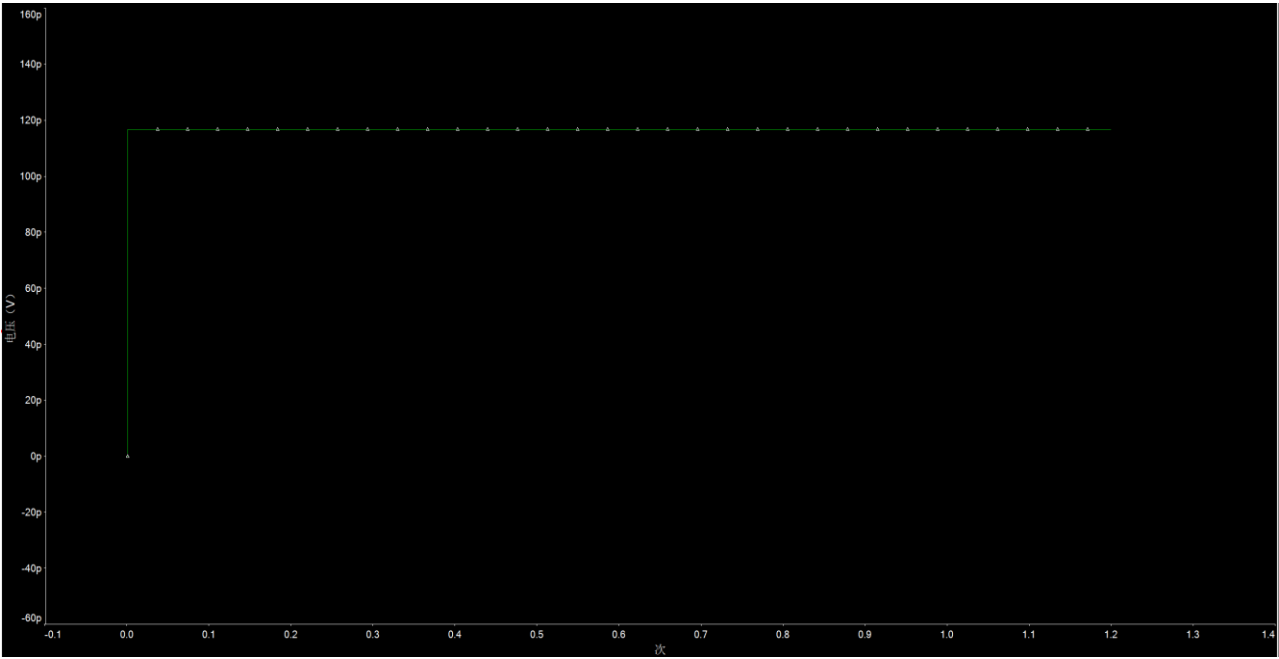


Figure 8. Current monitor is approximately 117 pV, corresponding to about 117 pA.
Note: Multisim simulation. V(3) is a current monitor: 1 V represents 1 A.

Shorted DC link

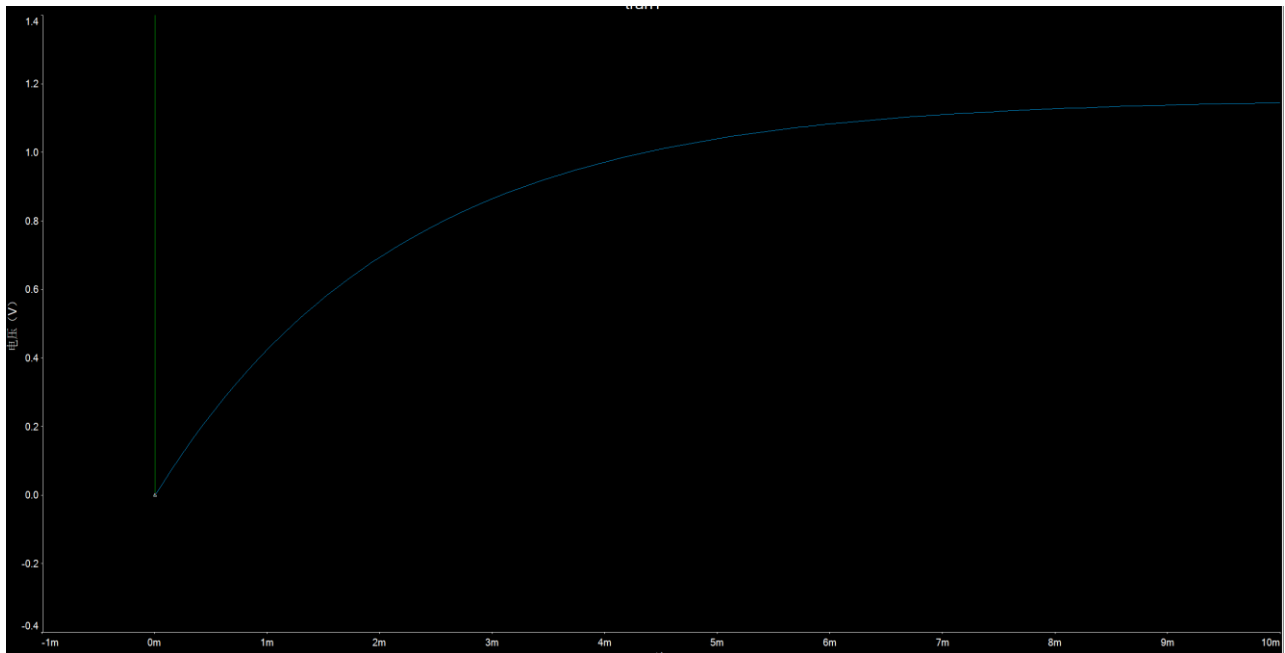


Figure 9. DC-link voltage rises to about 1.156 V over a few milliseconds.

Note: Multisim simulation. V(3) is a current monitor: 1 V represents 1 A.

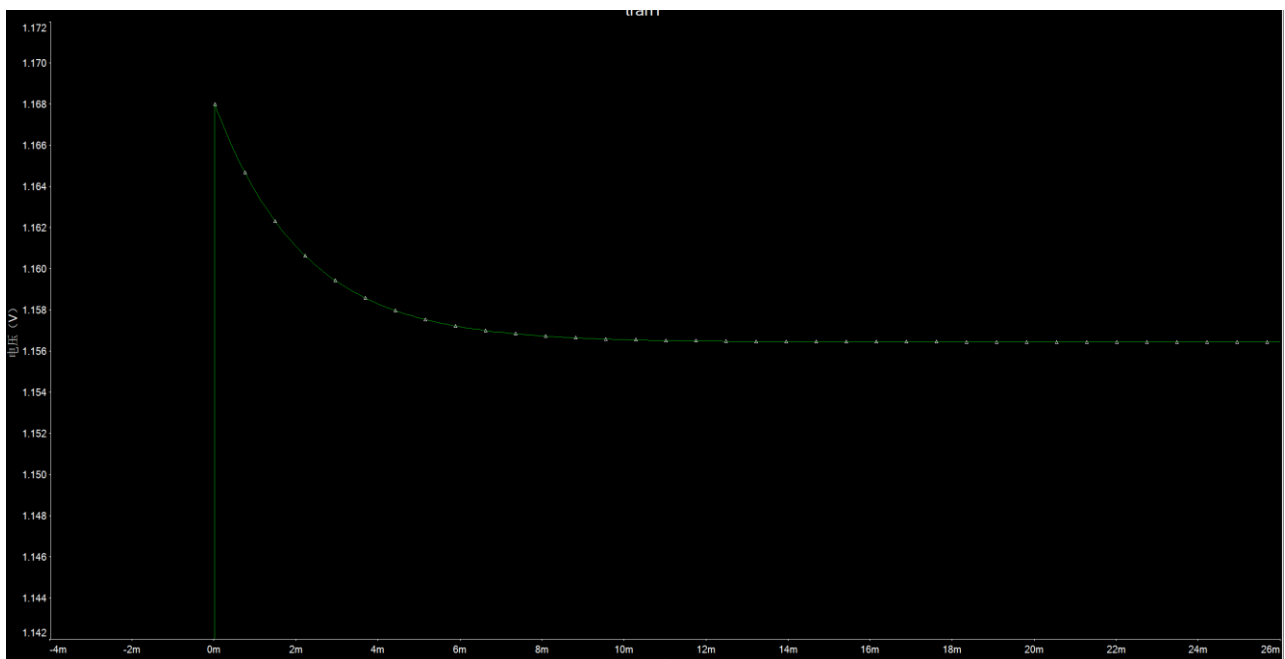


Figure 10. Current falls slightly from 1.168 A and persists near 1.156 A.

Note: Multisim simulation. V(3) is a current monitor: 1 V represents 1 A.

Signals and proposed response

I would measure pack voltage and DC-link voltage across the inverter input. I would also measure precharge current, relay commands and auxiliary feedback. A timer and sensor checks detect failed or implausible readings. Voltage and current together help distinguish an open path from a short. Voltage alone is not enough.

Condition	Proposed action
$V_{dc} \geq 0.95V_{pack}$; current decays; feedback agrees	Permit second AIR closure, then complete the precharge sequence.
Low bus and near-zero current	Abort; report an open-path/feed fault; inspect before retry.
Low bus and sustained high current	Abort promptly; open precharge relay and AIRs; prohibit automatic retry.
Voltage or contact feedback implausible	Reject startup and diagnose the sensor/contactactor fault.

The netlists show electrical fault signatures. They do not implement automatic cutoff, contactor sequencing, a thermal model or fault latching. The responses are proposed design requirements. Thresholds and timeout need tolerance checks, real inverter load data and team review. The modelled short gives about 133.73 W of sustained resistor power. Unlimited retry is unacceptable.

B.2 Accumulator safety concept

English concept word count: 514 (excluding heading and references).

The accumulator powers the car. My safety plan aims to prevent electric shock, fire, cell damage and unsafe operation. It covers driving, charging and maintenance. I use the brief's 32-series LFP pack, rated at 102.4 V nominal and 116.8 V maximum. It has an Orion BMS 2, a service disconnect and a pre-charge path. Before selecting parts, I would confirm the cell limits, fuse rating and AIR model with the CSG subsystem lead. Cooling, sensor locations and inverter leakage also need confirmation.

The first layer is preventative. A correctly rated DC fuse limits short-circuit damage. It must protect the weakest downstream conductor or component. Its interrupt rating must exceed the available fault current. FSUK v1 requires suitable over-current protection and a fused TS high-current path (EV3.2.1–EV3.2.7) [2]. AMS/BMS monitoring is detective. It identifies cell over-voltage, under-voltage and excessive temperature before damage becomes worse. EV5.8.3 requires continuous monitoring of cell voltage, TS current and temperature. Temperature sensors must cover at least 30% of lithium cells, evenly distributed through the accumulator [2]. BMS failure or loss of supply must not produce a healthy status. The shutdown system must fail safe and open the AIRs and pre-charge relay. AMS or IMD trips use non-programmable latching. A person at the vehicle other than the driver must reset them manually (EV6.1.6–EV6.1.9) [2].

An IMD detects loss of insulation between the TS and chassis. It opens the shutdown circuit.

Pre-charge is both preventative and detective. It limits current into the inverter capacitor. Measured bus voltage must reach the required level before the second AIR closes (EV5.7.1–EV5.7.2) [2]. If voltage fails to rise, the system stops rather than repeatedly retrying. I would compare AIR feedback with bus voltage to detect welded or failed-to-close contacts. Opening the SDC and isolating the pack are mitigating actions. EV6.1.5 requires immediate TS shutdown and interruption of accumulator current. Within five seconds, TS voltage must fall below 60 VDC and 25 VAC RMS [2].

A welded AIR may prevent isolation even after an open command. Before work, check residual voltage and use the maintenance disconnect.

For maintenance, a removable service plug separates accumulator segments. The TSMS must be lockable off. These prevent accidental re-energisation (EV5.4.3–EV5.4.4; EV6.2.2) [2]. Follow the team's lockout/tagout process and wait for stored energy to discharge. Prove the meter on a known source. Measure all relevant points, then prove the meter again. The vehicle-side indicator required by EV5.4.7–EV5.4.9 helps, but does not replace these measurements [2].

A protective enclosure and suitable insulation help prevent accidental contact and impact damage. Insulation must suit the voltage and temperature. Protected cable routing, secure connectors and strain relief reduce abrasion and loose connections (EV3.1 and EV4.5) [2]. The supplied theses show accumulator construction, safety and cooling methods [4, 5]. I would use them to guide checks, rather than copy their dimensions, materials or limits. I would review FSUK rules and current CSG decisions. Fault current, fuse coordination, insulation and thermal limits need fresh calculations. I would then seek a reviewed LV or simulation test plan. I would not energise or physically test an HV accumulator for this assessment.

Section C LV hardware

The driver and feedback diagrams below are hardware proposals. Simulation uses simplified models. It does not qualify the physical parts.

C.1 Contactor coil driver

I assume a nominal 12 V supply and check 10.8–14.4 V. $R_{coil} = 24 \Omega$ and $L_{coil} = 100 \text{ mH}$. Switching is no faster than 1 Hz. CMD12 is a ground-referenced 0/12 V push-pull output with at least 15 mA capacity. These assumptions need checking against the actual coil and controller. The SDC opens high-side coil supply independently of CMD12.

Figure 11 shows the driver, gate protection and flyback connections.

C.1 12 V coil: low-side drive and local turn-off clamp

Schematic proposal | filled dots = connected nets | A = anode; K = cathode

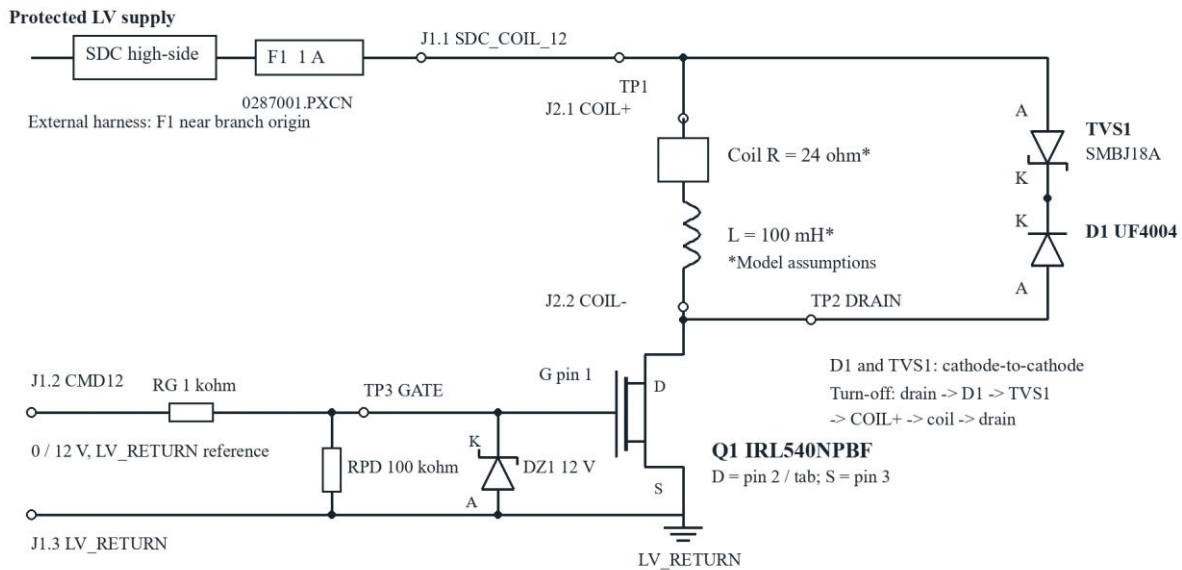


Figure 11. Proposed low-side coil driver and local turn-off clamp.

Note: hardware proposal; RCOIL/LCOIL are the assumed coil load. The simulations use functional models.

Design and Multisim model

Figure 11 shows my hardware driver proposal. Figure 12 shows the imported Multisim driver and feedback model. $R_{COIL} = 24 \Omega$ and $L_{COIL} = 100 \text{ mH}$ represent the coil. Sources, parasitic capacitances and controlled-source monitors support the tests. They are not all hardware BOM items. Figure 21 shows the separate protected feedback proposal.

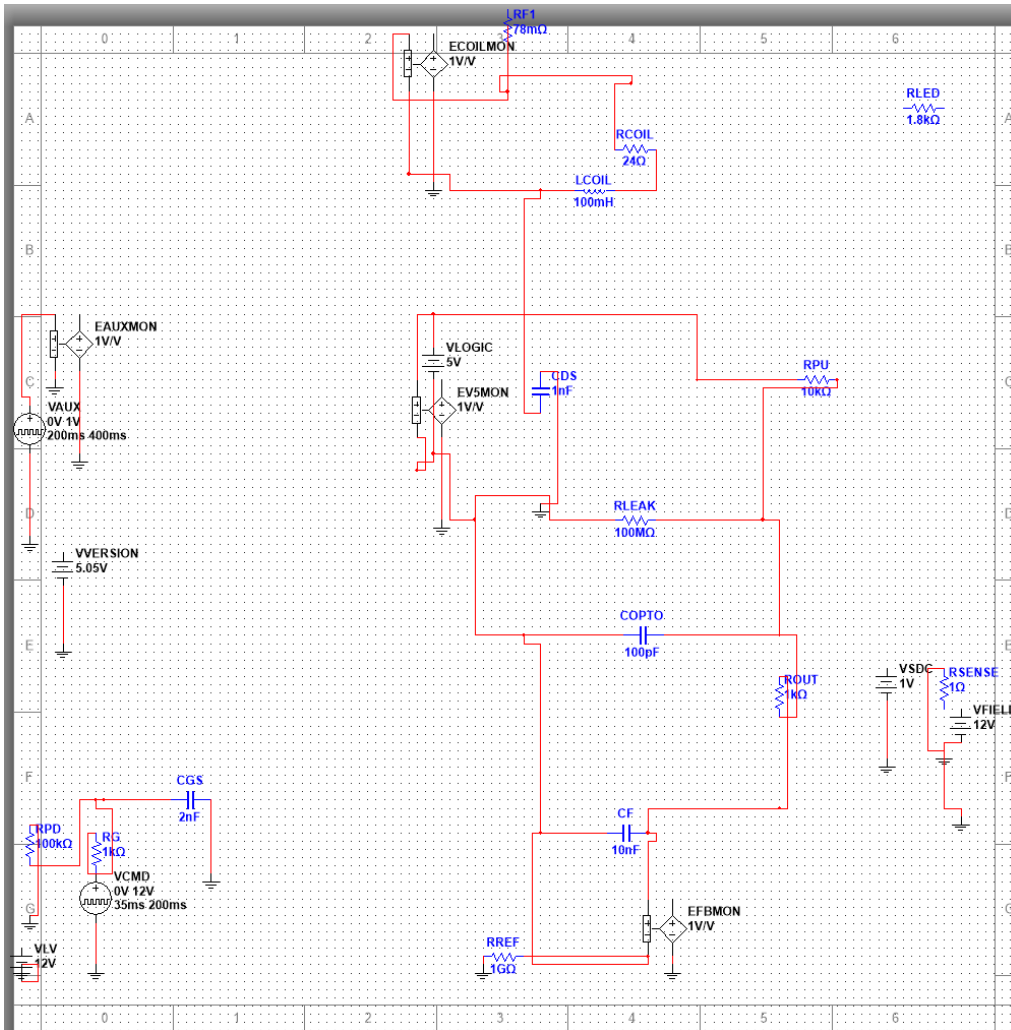


Figure 12. Multisim schematic view of the C driver and feedback simulation model.

Source: my Multisim model view. This is a partial screenshot, and RLED is unconnected. The supplied netlists contain the complete model definitions.

C.1 Selection, protection and calculations

When SDC is closed and CMD12 is high, Q1 connects the coil return to LV_RETURN. Current flows through F1, the coil and Q1. RPD discharges the gate when the command is removed or its wire opens. An open SDC removes coil power even if CMD12 stays high.

MOSFET and gate network

Q1 is the Infineon IRL540NPBF. VDS is rated at 100 V and the absolute VGS limit is ± 16 V. Maximum RDS(on) is 44 milliohm at VGS = 10 V. At 0.6 A, conduction loss is about $0.6^2 \times 0.044 = 15.8$ mW. Using twice that resistance gives about 32 mW. Switching loss is additional. The actual gate voltage must fully enhance Q1. [7]

RG = 1 kohm limits gate charging and abnormal input current. RPD = 100 kohm keeps Q1 off by default. DZ1 = BZX55C12-TAP has its cathode at the gate and anode at the source. It protects against positive and negative gate voltage. Its nominal rating is 12 V. At 5 mA and 25°C, the specified range is 11.4–12.7 V. Current and temperature change the clamp voltage. I would check that worst-case VGS stays within ± 16 V. [7, 11]

Flyback diode and TVS

D1 = UF4004-E3/54 and TVS1 = SMBJ18A sit across the coil. They are connected in series with their cathodes joined. TVS1 is unidirectional. At turn-off, drain current flows forward through D1, then through

TVS avalanche to COIL+. This gives more reverse coil voltage than a diode alone and speeds up current decay. Mechanical release time still needs checking for the actual contactor. [8, 9, 10]

Check	Result and basis
I, tau	12/24≈0.5 A; 14.4/24≈0.6 A; L/R=4.17 ms, neglecting series drops.
Ecoil	$\frac{1}{2}LI^2 \approx 12.5$ mJ at 0.5 A; 18 mJ at the assumed 0.6 A maximum.
VDS bound	SMBJ18A: 18 V standoff and 20–22.1 V breakdown. Maximum clamp is 29.2 V at the specified pulse current. Allowing 1.7 V for D1 gives 14.4+29.2+1.7=45.3 V<100 V. Layout overshoot is excluded. [9, 10]
Pulse stress	Conservative TVS peak: $\leq 29.2 \times 0.6 = 17.5$ W. Average total coil-energy bound: 18 mW at ≤ 1 Hz. The 600 W rating is for a 10/1000 μ s pulse. It is not continuous power or a fixed energy guarantee. Check actual pulse length and temperature derating. [9]

Fuse and fault boundary

F1 is 0287001.PXCN, with a 0FHA0001ZXJ holder. It is rated at 1 A and 32 V DC. Interrupting capacity is 1 kA at 32 V DC. I would place it near the start of the SDC coil branch. The assumed 0.6 A maximum is 60% of its rating. The fuse curve must suit the wiring, connectors, temperature and available short current. A Q1 drain-source short may draw only normal coil current. It may therefore leave F1 intact. The independent high-side SDC must remove power. [13]

I tested normal operation, SDC opening, a broken command wire and coil supply loss. The plots show electrical behaviour, not mechanical contact release. In the SDC test, I keep the command high to check independent shutdown.

Simulation signals are V(7) command, V(6) gate, I(LCOIL) coil current and V(5) drain. V(25) is feedback and V(26) is logic supply. V(27) measures differential coil voltage. V(28) is the prescribed auxiliary-contact state. V(100) marks the version. The V5 model uses behavioural switches, an assumed 20% effective CTR and preset contact timing. It does not model mechanical motion or full device behaviour.

Normal coil operation

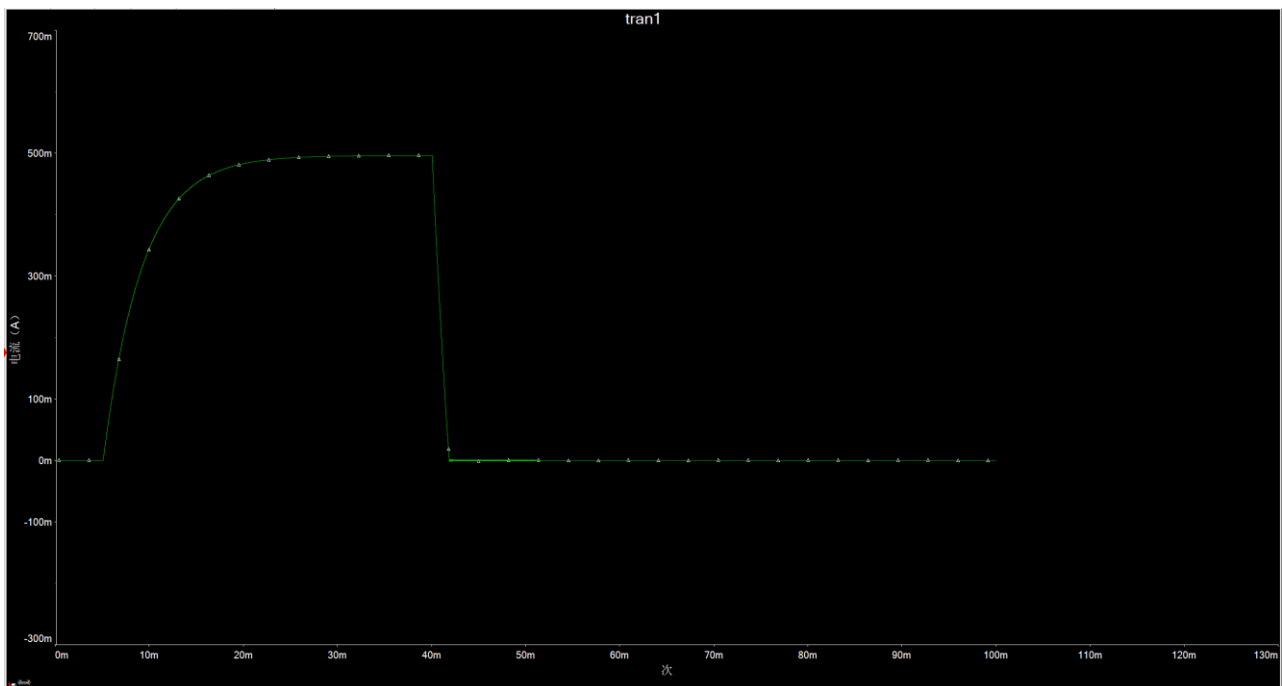


Figure 13. Normal coil current rises and decays after turn-off.

Note: Multisim simulation; m denotes milliseconds on time axes and milliamps on current axes.

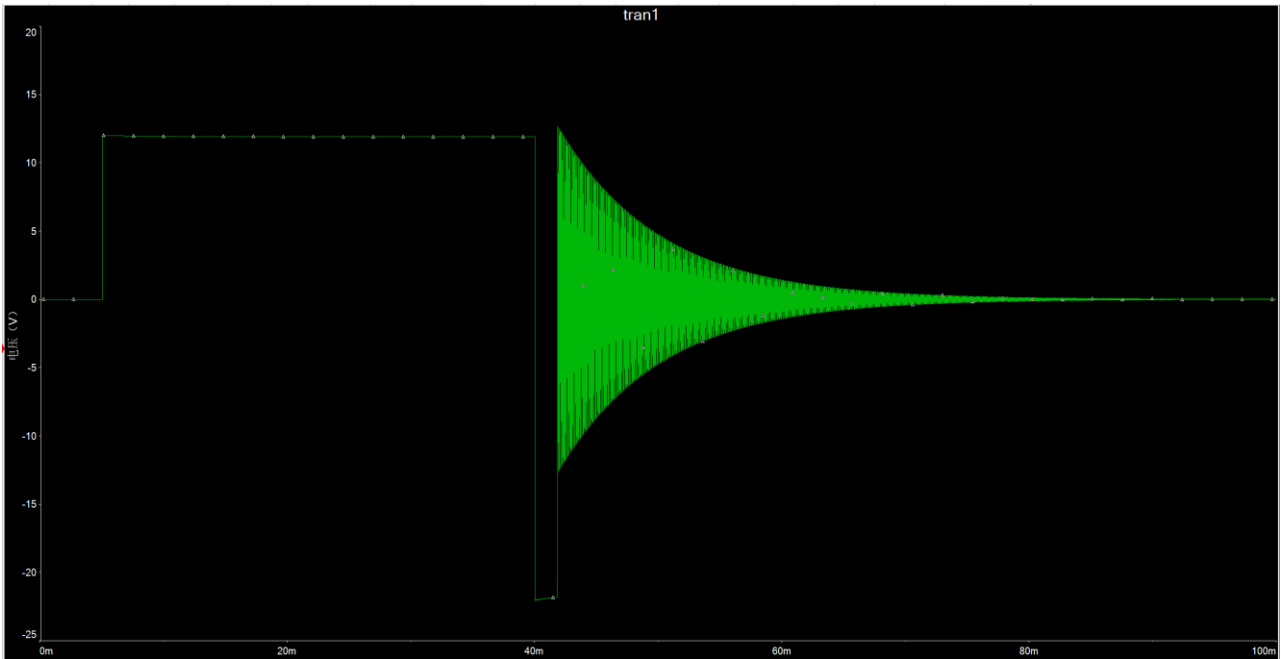


Figure 14. Coil voltage changes from positive energisation to a negative clamped turn-off voltage.

Note: Multisim simulation; m denotes milliseconds on time axes and milliamps on current axes.

Turn-off ringing depends on the idealised parts, drain capacitance and solver timestep. This does not prove that physical ringing is acceptable. About 2 ms of electrical current decay is not a measured mechanical release time.

Shutdown circuit opening

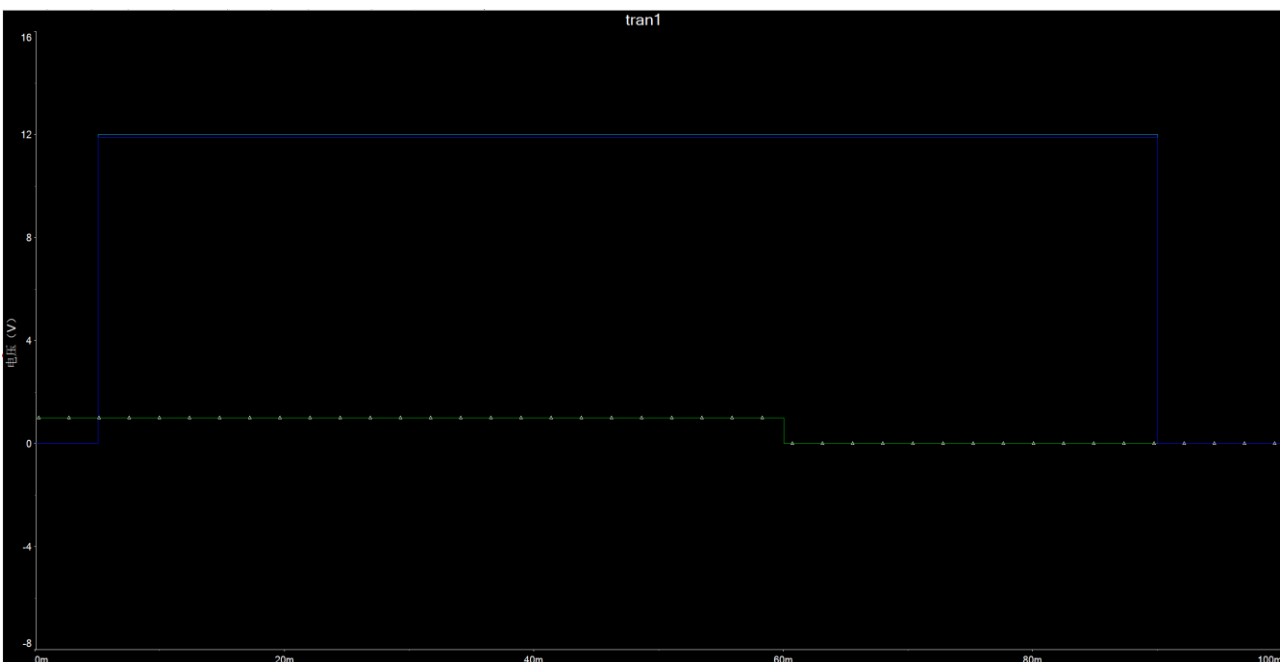


Figure 15. SDC model enable falls near 60 ms while command/gate remain high.

Note: Multisim simulation; m denotes milliseconds on time axes and milliamps on current axes.

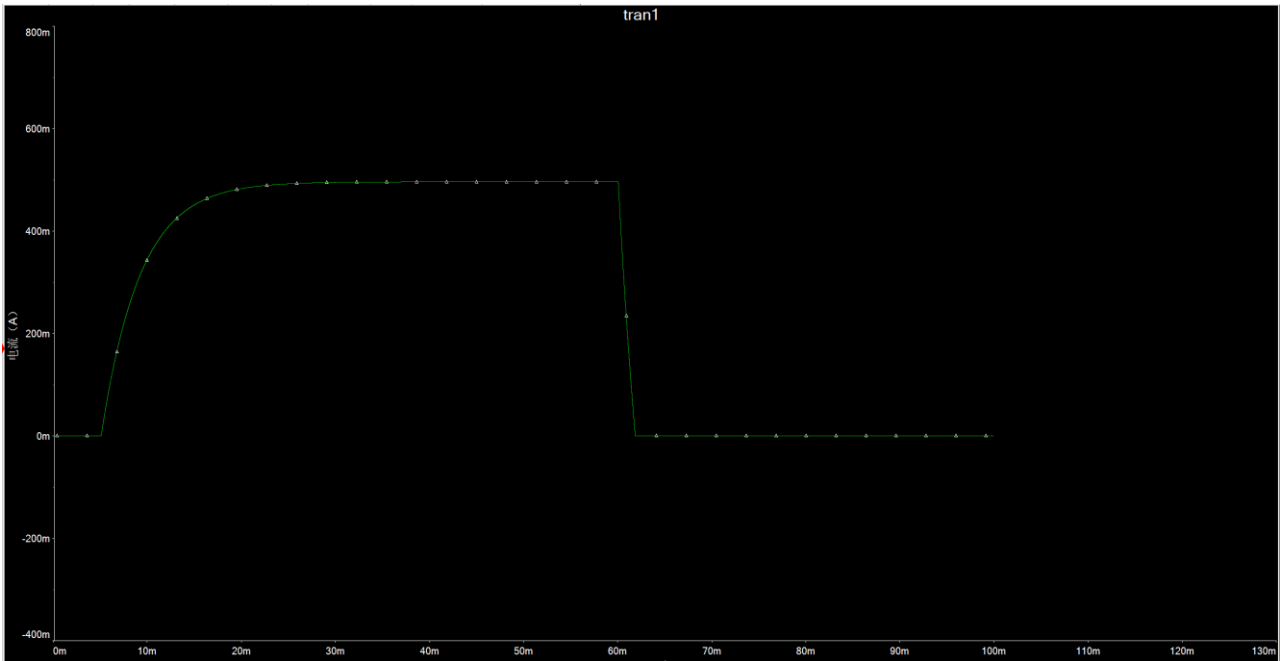


Figure 16. Coil current decays after the SDC opens despite the continuing command.

Note: Multisim simulation; m denotes milliseconds on time axes and milliamps on current axes.

Command wire opening

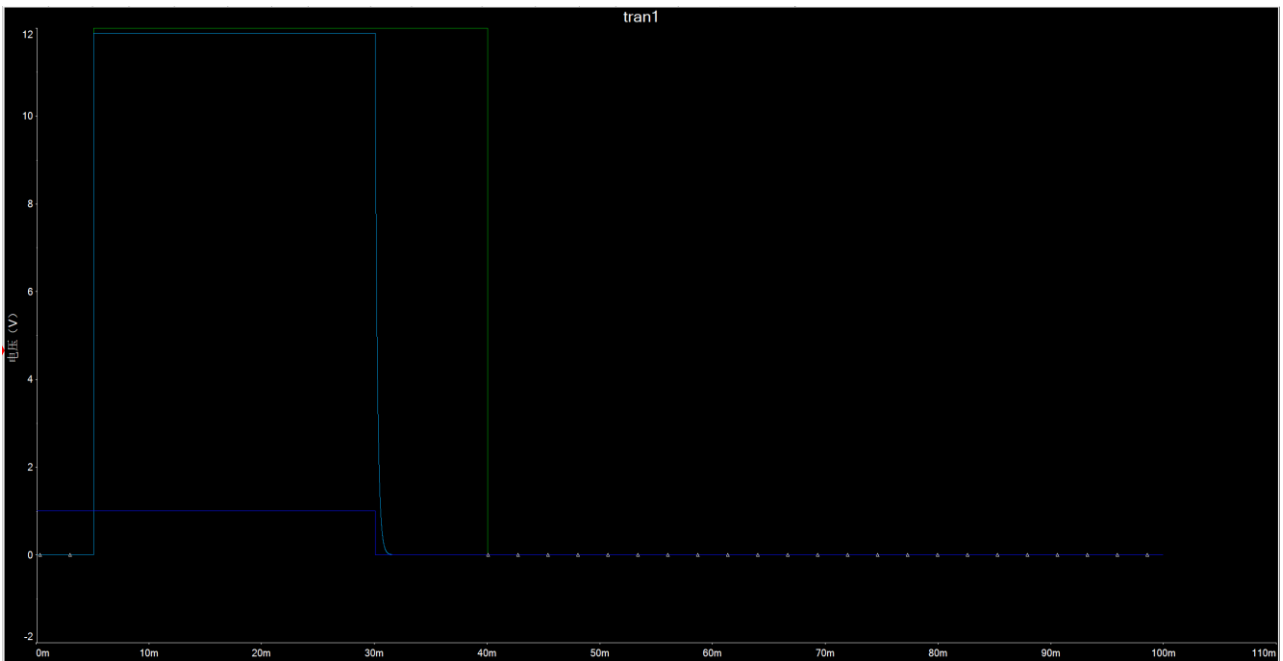


Figure 17. Controller command persists while the broken command connection causes the local gate to fall.

Note: Multisim simulation; m denotes milliseconds on time axes and milliamps on current axes.

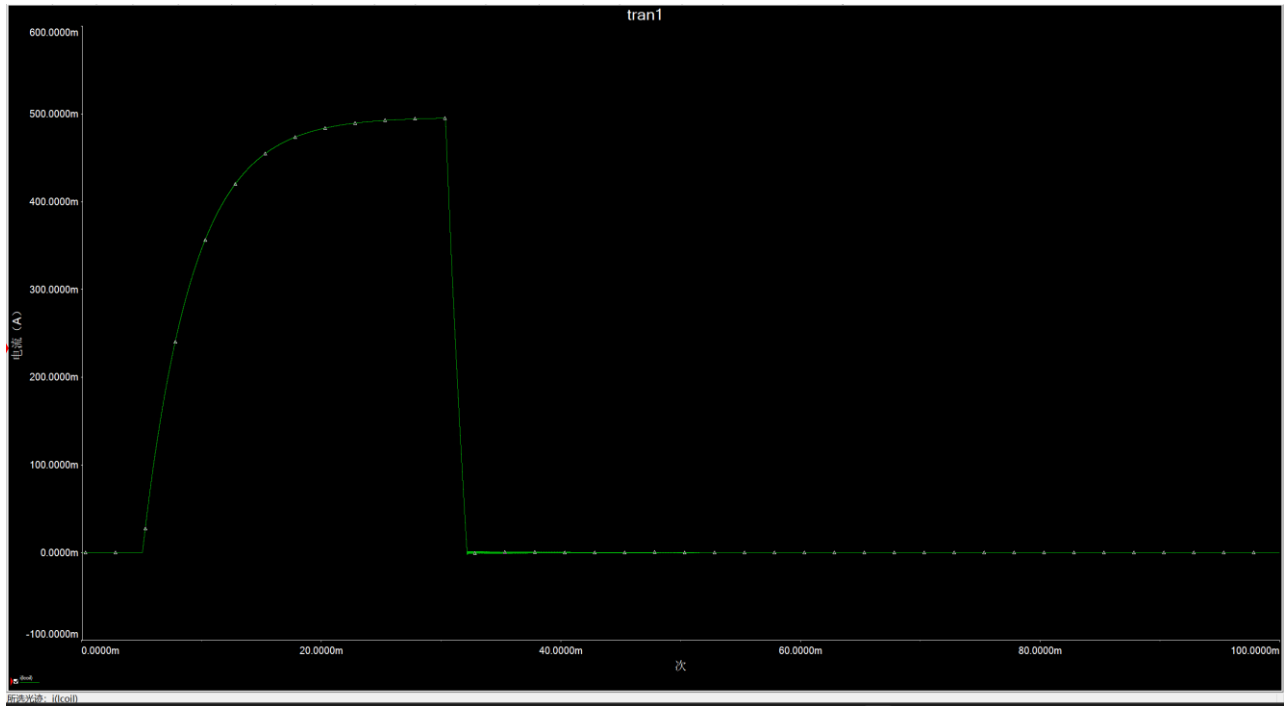


Figure 18. Coil current decays after the command wire opens.

Note: Multisim simulation; m denotes milliseconds on time axes and milliamps on current axes.

Loss of coil supply

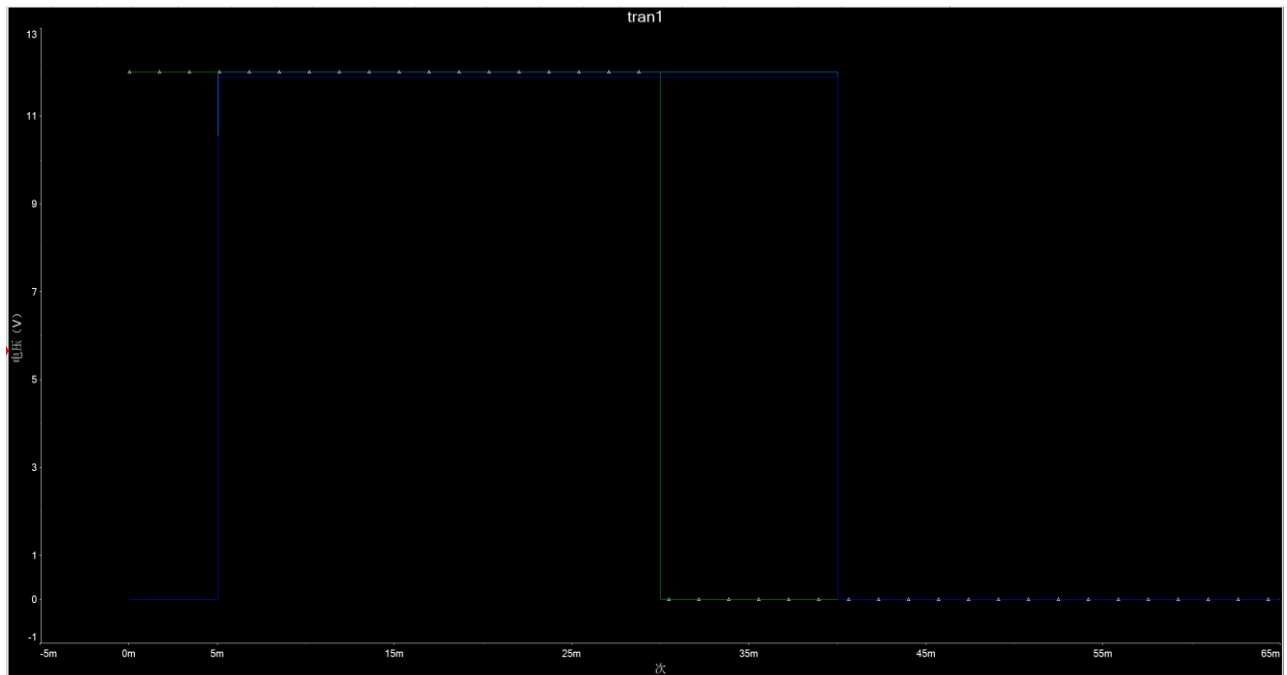


Figure 19. Coil supply is removed while command initially persists.

Note: Multisim simulation; m denotes milliseconds on time axes and milliamps on current axes.

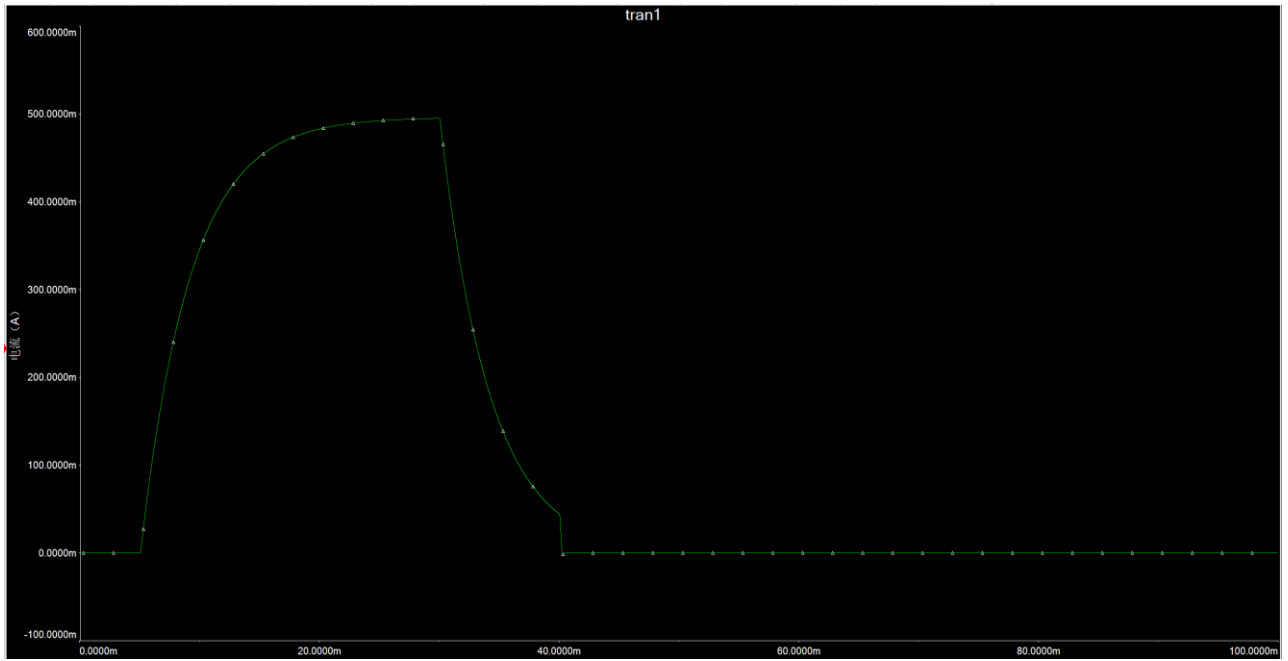


Figure 20. Coil current decays following supply loss.

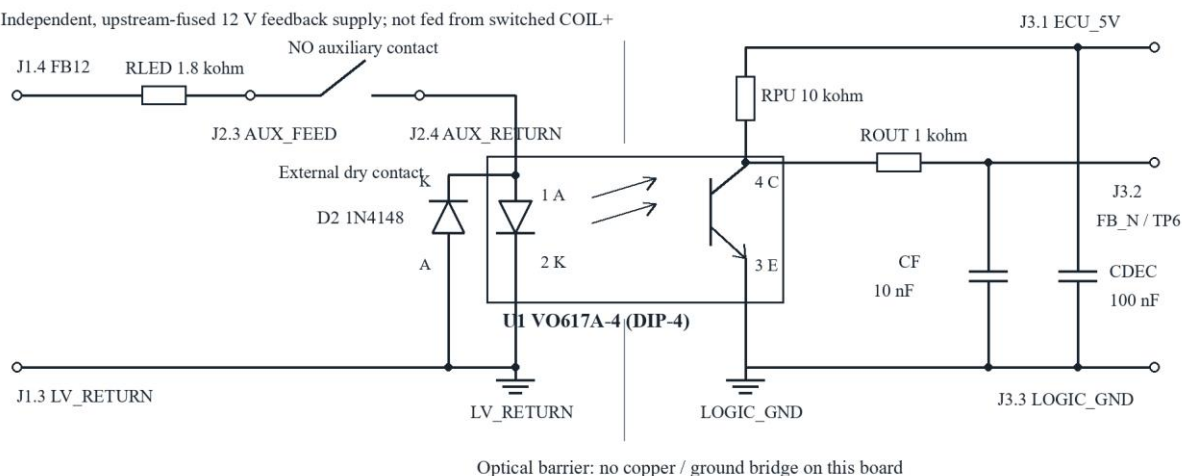
Note: Multisim simulation; m denotes milliseconds on time axes and milliamps on current axes.

C.2 Status feedback and input protection

Figure 21 shows the independent auxiliary-contact input and optocoupler output.

C.2 Protected auxiliary-contact feedback to 5 V logic

Independent, upstream-fused 12 V feedback supply; not fed from switched COIL+



AUX closed: LED on -> transistor on -> FB_N LOW. AUX open / field wire open: FB_N HIGH.

RLED is placed before the external contact to limit current in both field wires. D2 protects LED reversal.

True end-to-end galvanic isolation requires the two grounds to remain unbonded elsewhere.

J1.5 unused. Connector cavity numbering is defined in the pinout table, not by drawing left-to-right order.

Figure 21. Proposed auxiliary-contact feedback and protected 5 V input.

Note: FB12 is independently fused upstream of SDC. LV_RETURN and LOGIC_GND remain separate.

Feedback reads an independent auxiliary contact. CMD12 and drain voltage do not replace contact-state feedback. RLED sits before the external contact and limits current in both field wires. D2 is antiparallel to the input LED. The output pulls up only to ECU 5 V. A separate upstream-fused supply keeps feedback available after SDC opens.

Design check	Explanation
RLED / IF	RLED is $1.8 \text{ kohm} \pm 1\%$. Nominal $IF \approx (12 - 1.3) / 1800 = 5.94 \text{ mA}$. With a conservative 1.65 V input drop, $IF = 5.03 \text{ mA}$ at 10.8 V. With an assumed 0.8 V drop, $IF = 7.63 \text{ mA}$ at 14.4 V. A shorted LED gives resistor loss $\leq 0.117 \text{ W}$. Select 0.6 W. [12]
U1 / CTR	VO617A-4 CTR is 160–320% at $IF = 5 \text{ mA}$, $VCE = 5 \text{ V}$ and 25°C . I assume 20% effective CTR in simulation. At 5 mA LED current, this gives about 1 mA against a 0.5 mA pull-up demand. Check the assumption for temperature, ageing and saturation. [12] (project simulation evidence)
RPU / ROUT / CF	A 10 kohm pull-up gives default HIGH. A 1 kohm series resistor and 10 nF to logic ground filter the input. RC scales are about 10 μs falling and 110 μs rising. These are not mechanical debounce or diagnostic timeout values. Add 100 nF of local 5 V decoupling.

Keep LV_RETURN and LOGIC_GND separate on this board. Bonding them elsewhere defeats end-to-end galvanic isolation. The optocoupler still separates the signal voltages. This is an LV feedback proposal, not a qualified TS insulation boundary.

C2 Multisim model view

Figure 22 shows the standalone C.2 model in Multisim. Figure 21 is the matching hardware proposal. VLOGIC supplies 5 V. RPU, ROUT and CF model the pull-up and output filter. EFBMON and EV5MON monitor feedback and logic supply for plotting. COPTO, RLEAK and RREF are modelling aids, not extra hardware BOM items.

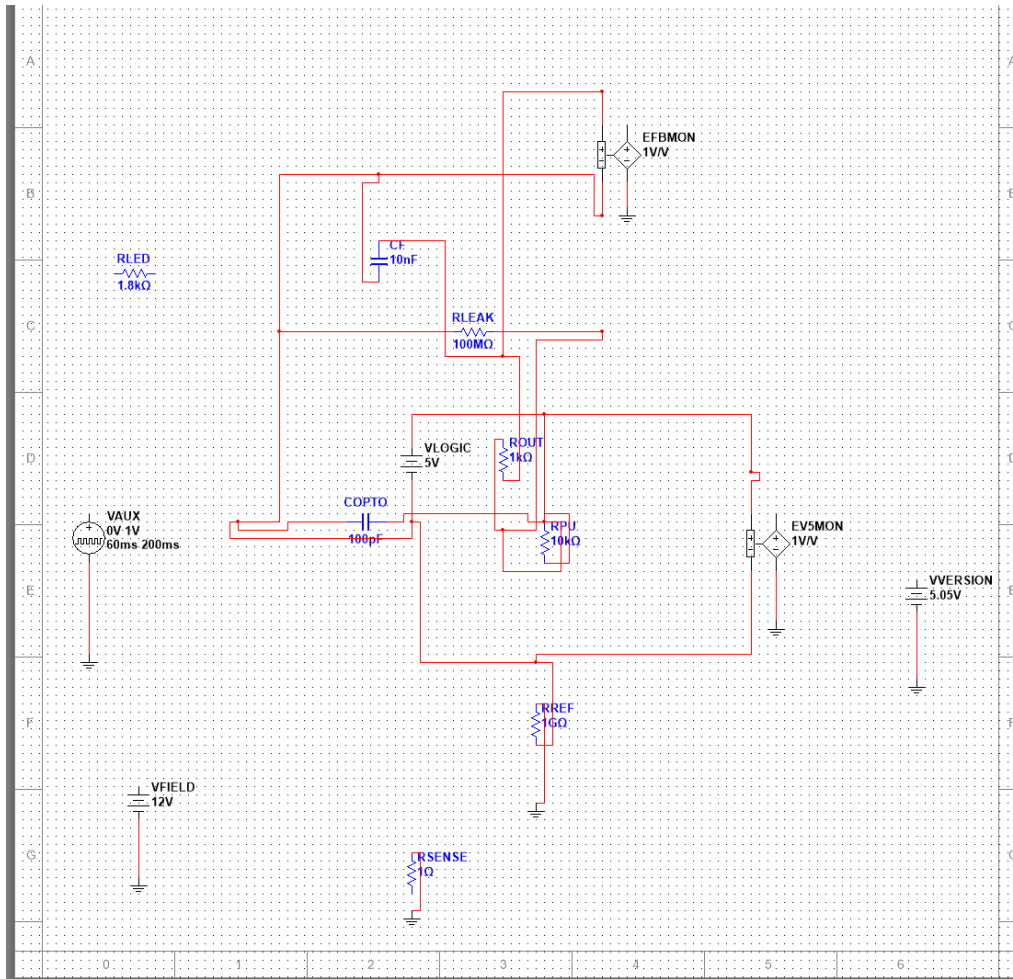


Figure 22. Multisim schematic view of the standalone C2 feedback model.

Source: my C2 Multisim model view. Some behavioural elements are not shown. RLED and VFIELD are unconnected in this view. C2_V5_Feedback.cir defines the functional test model.

Interfaces, test points and fault interpretation

Use different pole counts for J1/J2 JST VH connectors. Use JST XH for J3. Find pin 1 from the manufacturer drawing and moulded mark. Do not guess the viewing direction. AWG22 is a preliminary coil-wire choice and needs installation derating. The VH current rating depends on wire size. [14, 15]

Pin	Net	Function
J1.1	SDC_COIL_12	Coil supply after SDC and F1
J1.2	CMD12	0/12 V controller command
J1.3	LV_RETURN	Coil / LED-side reference
J1.4	FB12	Independent, upstream-fused feedback supply
J1.5	NC	Unused; no terminal fitted
J2.1 / J2.2	COIL+ / COIL-	Coil positive / Q1 drain
J2.3 / J2.4	AUX_FEED / AUX_RETURN	Two ends of the normally-open auxiliary contact
J3.1	ECU_5V	Logic-side 5 V

Pin	Net	Function
J3.2	FB_N	Filtered status; LOW = auxiliary contact closed
J3.3	LOGIC_GND	ECU logic reference

Test point definitions

Test point	Measurement and expectation
TP1 / TP2	COIL+ and DRAIN relative to LV_RETURN; their difference is coil voltage, which can reverse during turn-off.
TP3	GATE relative to SOURCE/LV_RETURN; verify off-state and VGS overshoot.
TP4 = J1.4	FB12 relative to LV_RETURN; should remain available after SDC opening.
TP5 = J3.1	ECU_5V relative to LOGIC_GND; approximately 5 V.
TP6 = J3.2	FB_N relative to LOGIC_GND; open ≈ 5 V, closed LOW.

Diagnostic and layout principles

HIGH means contact closure is unconfirmed. An open contact, broken wire or lost field supply can all cause this. LOW after command removal may indicate a mismatch. Wait for the verified release and debounce deadline before deciding. An output short to ground can produce the same LOW reading. Inhibit restart and request shutdown through an independent path. Auxiliary feedback alone does not prove main-contact welding. Set deadlines from the actual contactor. This model does not implement the diagnosis.

Place the clamp loop close to the coil connector. Keep the pulldown and zener close to Q1. Route coil-current return away from the gate and feedback signals. Maintain optocoupler separation, check crimp pull strength and provide strain relief. The listed connectors suit an enclosed LV prototype. Vehicle sealing and vibration suitability need team-specific checks.

I tested an open feedback contact and a prescribed stuck-closed contact. Feedback is HIGH for the open input and LOW for the held-closed contact.

Open feedback contact

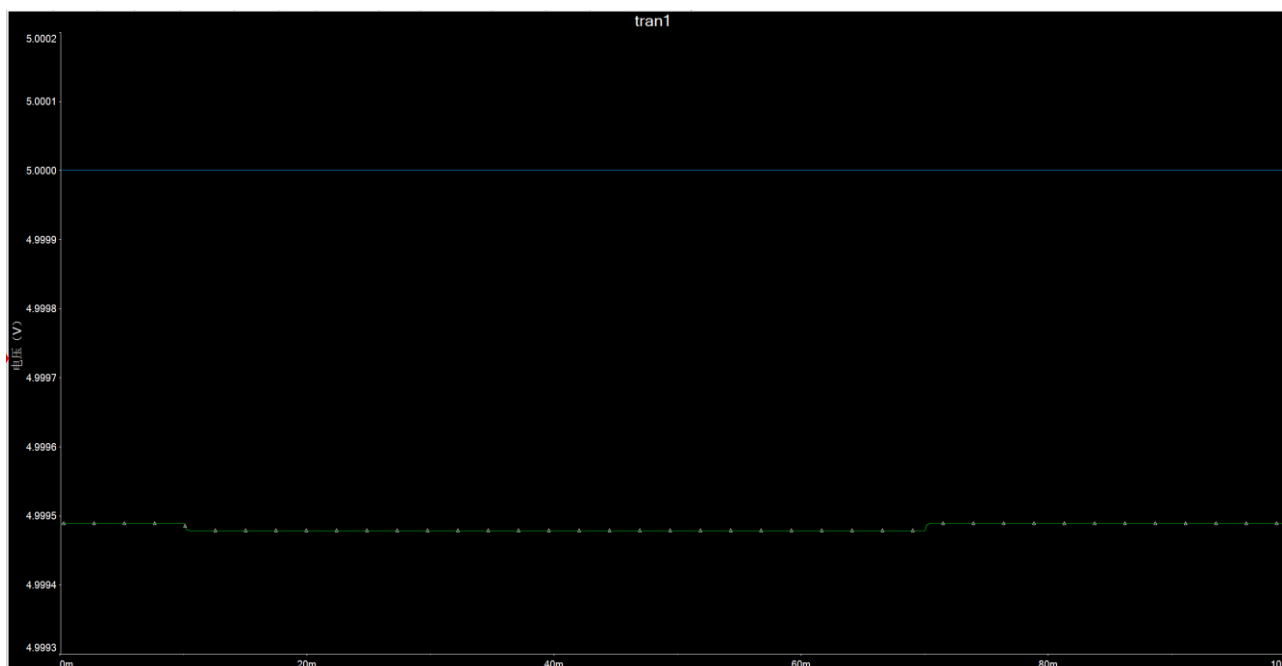


Figure 23. Open feedback produces an unconfirmed/high logic status.

Note: Multisim simulation; m denotes milliseconds on time axes and milliamps on current axes.

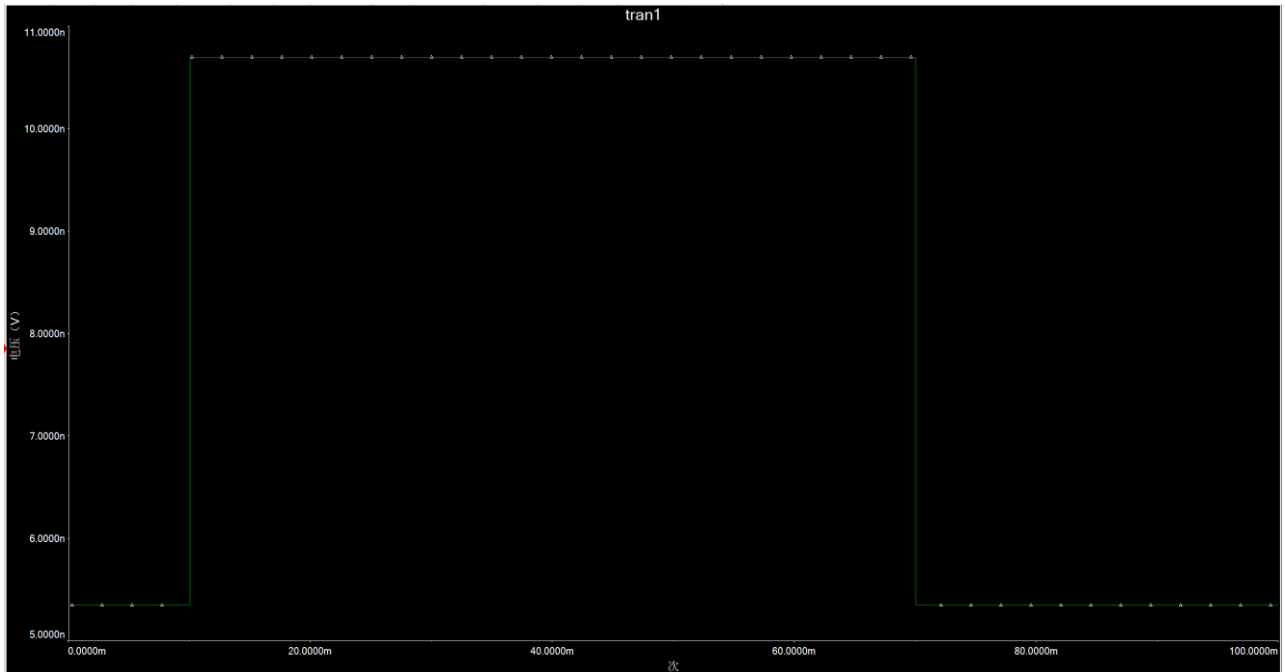


Figure 24. Feedback LED current is near zero in the open case.

Note: Multisim simulation; m denotes milliseconds on time axes and milliamps on current axes.

Stuck auxiliary feedback

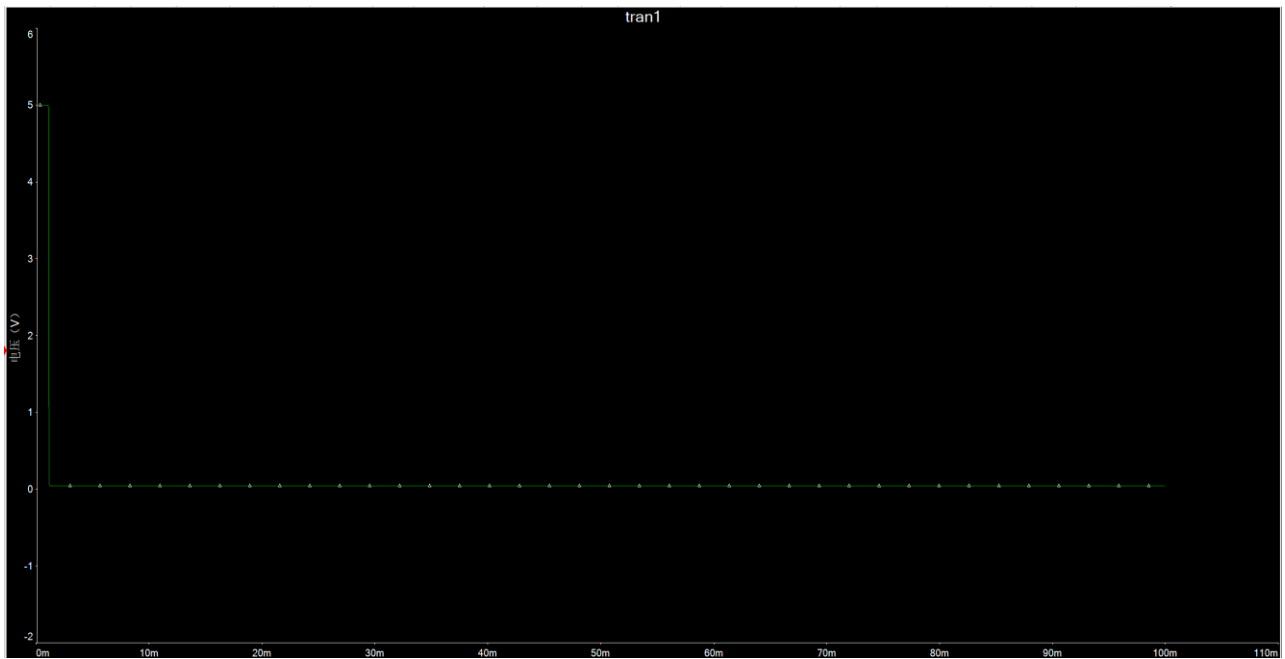


Figure 25. The prescribed stuck auxiliary contact holds feedback low.

Note: Multisim simulation; m denotes milliseconds on time axes and milliamps on current axes.

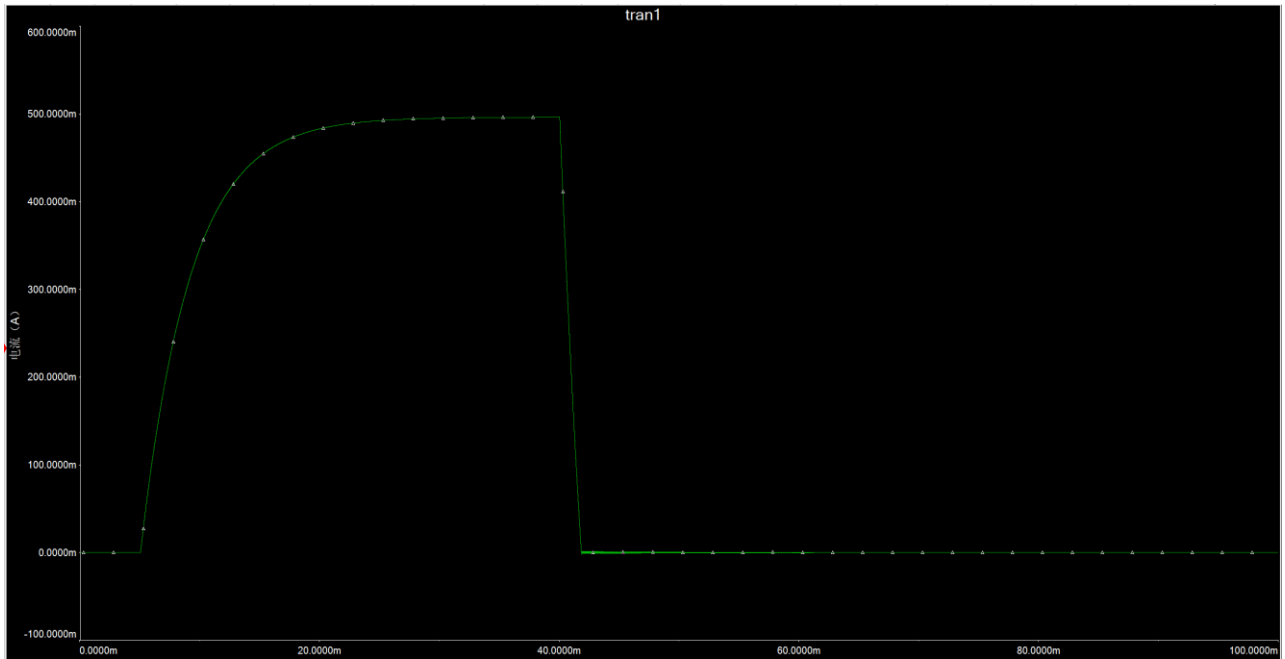


Figure 26. Coil current still decays after command removal.

Note: Multisim simulation; m denotes milliseconds on time axes and milliamps on current axes.

After command removal at 40 ms, coil current decays. The prescribed stuck contact keeps feedback LOW. This is a feedback mismatch. It does not prove that the main contacts have mechanically welded.

Proposed hardware acceptance

On a current-limited LV bench, first check fusing, pinout and polarity. Check that the two grounds are not accidentally bridged. Disconnect the command and confirm default-off. Use differential or isolated measurements so scope grounds do not bridge the domains. At 10.8, 12 and 14.4 V, measure current, VGS and VDS peak. Check clamp duration and repeated-operation temperature rise. Keep the command high while testing SDC shutdown.

Measure actual auxiliary-contact delay and bounce. Test broken-wire, stuck-contact and lost-5 V diagnosis. Before buying parts, check whether the coil has a suppressor or economiser. Confirm permitted clamp voltage, cold resistance and maximum inductance. Check auxiliary-contact ratings and wetting current. Also check fuse and harness coordination. These are proposed acceptance tests, not completed physical tests.

Completed simulation work

The model covers normal operation and five fault cases. Hardware release also requires fuse, transient/EMC and mechanical contact checks.

The component list and buying links are in Appendix B and components/C_BOM.csv. Simulation assumes $0.078\ \Omega$ fuse resistance. The selected fuse's typical cold resistance is $0.123\ \Omega$. RREF, RSENSE and monitor sources are simulation aids. They are not purchased hardware or intended isolation connections. CDEC is a hardware decoupling recommendation. The original plots do not validate it.

Section D Embedded task

D.1 Home battery monitoring

This task is for home storage, not the vehicle accumulator. I use the supplied mock interfaces to read controller data and raw inputs. I check BMS and inverter communication and data age. I check battery temperature and relay feedback directly. A door/tamper event requests shutdown. Zero PV output can be normal at night, so it does not cause shutdown. Relay command, buzzer and notification are outputs. No real bus protocol is modelled.

I wrote the checks in `src/logic.c`. I tested them with the supplied native scenarios and in Wokwi. Appendix A contains the code and native results. Cell and grid limits are exercise assumptions. So is island operation. The code evaluates each snapshot without a persistent fault latch.

Communication and input design

BMS and inverter messages are periodic controller snapshots. Each has a communication flag and last-update timestamp. The ECU rejects missing messages or data older than 1,000 ms. Cell voltage, battery temperature and inverter status are safety-critical. Invalid data could hide an unsafe condition. The separate temperature sensor is a periodic raw input with a validity flag. Its mock interface has no timestamp. A real implementation would also need an acquisition timeout.

PV voltage and current are periodic raw readings for monitoring. Zero generation alone is not a fault. Grid voltage, frequency and presence are also periodic raw inputs. They determine whether grid connection is safe. Relay command is an output; contact feedback is an input. A disagreement is safety-critical. The door/tamper input is sampled at each scenario boundary. A trigger requests shutdown. Buzzer and notification are output commands. They report shutdown but do not replace the relay-opening request. Mock read/write functions represent these interfaces. They do not model a real bus or physical relay response.

Shutdown summary

The controller keeps the system ON only when the safety-related readings are available and plausible. It switches the system OFF if BMS or inverter communications fail, their data is more than 1,000 ms old, any cell voltage is below 3,000 mV or above 4,200 mV, either battery temperature reading reaches 60 °C, or the inverter reports a fault or is not running. It also switches OFF if the door/tamper input is triggered, the relay feedback disagrees with its command, the grid is present outside 207–253 V or 49–51 Hz, or the grid is absent while the relay is commanded or reported closed. For a shutdown, the logic requests the grid relay to open and turns on the buzzer and notification flag.

If the grid is absent and the relay is confirmed open, the system stays ON without an alarm. This assumes the inverter can operate safely in island mode. Zero PV output is not treated as a fault because it can occur normally, such as at night. Pack current is checked for a valid numeric value, but no current limit is applied because the task gives no pack rating or limit.

All numerical limits are assumptions for this software exercise, not certified battery or grid protection settings. Real limits must come from the battery, BMS, inverter and applicable grid requirements. The provided relay interface is a mock; this code does not prove that physical contacts open. AI assistance was used to draft and review the ``data_is_stale()`` and ``evaluate_system_state()`` functions. AI was also used to draft this summary.

The standalone SUMMARY.md is included unchanged. Its statement of AI assistance is retained. AI also assisted with combining and formatting this report and organising the supporting files.

Wokwi simulation check

I ran the same logic and all 15 supplied scenarios on a simulated Raspberry Pi Pico. Each scenario lasts about two seconds, then the sequence repeats. Green means SYSTEM ON. Red means SYSTEM OFF. Yellow means notification enabled. GP4 controls the buzzer output. GP6 reads the active-low door/tamper button.

Figure 27 shows the repeating scenario results. The visible normal-operation and grid-absent cases return ON with no alarm. The visible fault cases return OFF with buzzer and notification enabled. Grid absence is allowed only when the mock grid relay is open. This uses the island-mode assumption above. The red and yellow LEDs match the displayed fault output.

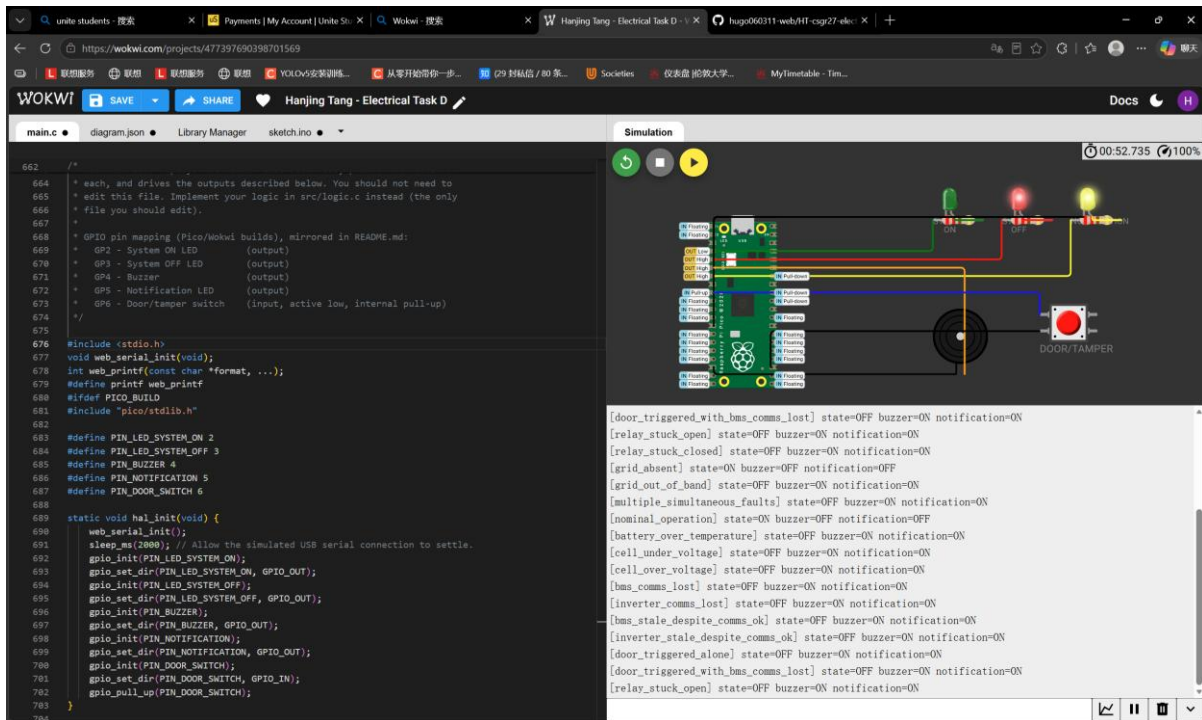


Figure 27 Wokwi scenario playback and GPIO outputs.

Source: my Wokwi simulation; serial results are shown beside the virtual circuit.

In the button test, nominal_operation changes from ON with no alarm to OFF with buzzer and notification enabled (Figure 28). The door/tamper input therefore overrides otherwise normal readings. The screenshot shows the shutdown transition.

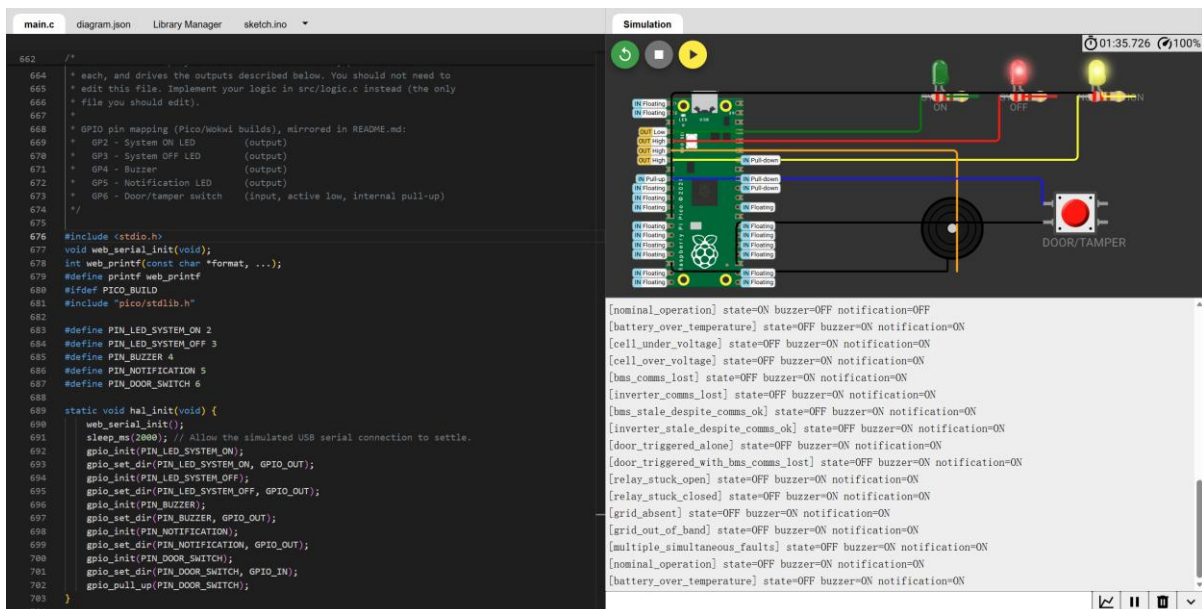


Figure 28 Door/tamper input changes nominal operation from ON to OFF.

Source: my Wokwi simulation; serial results are shown beside the virtual circuit.

Project links

GitHub repository: <https://github.com/hugo060311-web/HT-csgr27-electrical-task>

Wokwi simulation: <https://wokwi.com/projects/477397690398701569>

External references

The references below are the assessment, rules, papers and manufacturer documents used in my answers. My diagrams and simulation plots appear in the main sections. Appendices A–C contain code, test records, component links and reproduction notes.

[1] CSG Racing. Electrical (LV & Embedded Systems) Assessment. Supplied brief, common design assumptions and Sections A–D.

[2] Institution of Mechanical Engineers. Formula Student UK 2026 Rules v1. Rule clauses cited in the text; the assessment specifies this version. Source

[3] Badawy, A. D., et al. Low-Voltage Control Circuits of Formula Student Electric Racing Cars. *Hardware*, 2024, 2(3), 190–222. DOI: 10.3390/hardware2030010. Figure 11 and Sections 2.5.1 and 5.3. CC BY 4.0. Source

[4] Räisänen, A. High Voltage Accumulator System for a Formula Student Vehicle. Bachelor's thesis, Metropolia University of Applied Sciences, 2023. Source

[5] Sanchez, D. M. High Density Accumulator Design for an Electric Vehicle. Instituto Superior Técnico, Universidade de Lisboa, July 2024. Supplied extended abstract.

[6] Ewert Energy Systems. Orion BMS 2 Wiring Manual. Digital output characteristics. Source

[7] Infineon. IRL540N / IRL540NPbF datasheet. Maximum ratings, electrical characteristics and pinout. Source

[8] TE Connectivity. Relay coil suppression with DC relays. Release implications of diode and zener suppression. Source

[9] Littelfuse. SMBJ Series TVS Diodes datasheet. Pulse ratings and derating. Source

[10] Vishay. UF4001–UF4007 datasheet, document 88755. Source

[11] Vishay. BZX55 Series datasheet, document 85604. Source

[12] Vishay. VO617A optocoupler datasheet, document 83430. Source

[13] Littelfuse. 287 ATOF Fuse datasheet. 32 V DC rating, interrupt rating and time-current characteristics. Source

[14] JST. VH connector catalogue. Ratings, contact/housing options and dimensions. Source

[15] JST. XH connector catalogue. Ratings, contact/housing options and dimensions. Source

[16] Vishay Sfernice. LTO 100 H datasheet, document 50081, revision 14 October 2022. Pulse and thermal selection information. Source

Appendix A Code and native test record

The following is an unchanged copy of src/logic.c. The supplied types, mock interfaces and scenarios remain in include/, src/mock_hw.c and test/. The native output comes from this same source. SUMMARY.md is unchanged.

Listing D1 Implementation

```
// This is the only file you should edit.
// AI assistance: helped draft and review these fault checks; thresholds and
// behaviour still need to be understood and confirmed by the applicant.

#include <math.h>

#include "logic.h"
#include "mock_hw.h"

// These are exercise assumptions. A real battery must use limits from its BMS
// and battery manufacturer.
#define MIN_CELL_MV 3000.0f
#define MAX_CELL_MV 4200.0f
#define MAX_BATTERY_TEMP_C 60.0f
#define MAX_DATA_AGE_MS 1000u

static bool data_is_stale(uint32_t now_ms, uint32_t update_ms) {
    return (uint32_t)(now_ms - update_ms) > MAX_DATA_AGE_MS;
}

SystemOutputs evaluate_system_state(BmsData bms, BatteryTempData batt_temp,
                                   InverterData inverter, PvData pv,
                                   GridData grid, RelayState relay,
                                   DoorSwitchData door) {
    SystemOutputs out = {0};
    bool fault = false;
    uint32_t now_ms = read_system_time_ms();

    (void)pv; // Zero solar output can be normal, for example at night.

    // Missing or old data from either safety-related controller is unsafe.
    if (!bms.comms_ok || data_is_stale(now_ms, bms.last_update_ms) ||
        !inverter.comms_ok || data_is_stale(now_ms, inverter.last_update_ms)) {
        fault = true;
    }

    if (!batt_temp.valid || !isfinite(batt_temp.temp_c) ||
        batt_temp.temp_c >= MAX_BATTERY_TEMP_C ||
        !isfinite(bms.pack_temp_c) || bms.pack_temp_c >= MAX_BATTERY_TEMP_C ||
        !isfinite(bms.pack_current_a)) {
        fault = true;
    }

    for (int i = 0; i < BMS_NUM_CELLS; i++) {
        if (!isfinite(bms.cell_voltages_mv[i]) ||
            bms.cell_voltages_mv[i] < MIN_CELL_MV ||
            bms.cell_voltages_mv[i] > MAX_CELL_MV) {
            fault = true;
        }
    }

    if (!inverter.running || inverter.fault_code != 0) {
        fault = true;
    }

    // The grid limits are illustrative for this task, not certified settings.
    if (grid.present &&
        (!isfinite(grid.voltage_v) || !isfinite(grid.frequency_hz) ||
         grid.voltage_v < 207.0f || grid.voltage_v > 253.0f ||
```

```

    grid.frequency_hz < 49.0f || grid.frequency_hz > 51.0f)) {
    fault = true;
}

// Do not accept a closed grid connection when the grid is absent.
if ((!grid.present && (relay.commanded_closed || relay.feedback_closed)) ||
    relay.commanded_closed != relay.feedback_closed) {
    fault = true;
}

// Treat a tamper event as a shutdown request.
if (door.triggered) {
    fault = true;
}

if (fault) {
    out.state = SYSTEM_OFF;
    out.buzzer_on = true;
    out.notification_flag = true;
    write_relay(false);
} else {
    out.state = SYSTEM_ON;
}

return out;
}

```

Appendix A Native test output

Run the packaged source with `NATIVE_BUILD=ON`. The table shows software outputs for the supplied scenarios. It does not show physical relay operation. In `grid_absent`, the current source returns ON with buzzer and notification OFF.

Scenario	State	Buzzer	Notification
nominal_operation	ON	OFF	OFF
battery_over_temperature	OFF	ON	ON
cell_under_voltage	OFF	ON	ON
cell_over_voltage	OFF	ON	ON
bms_comms_lost	OFF	ON	ON
inverter_comms_lost	OFF	ON	ON
bms_stale_despite_comms_ok	OFF	ON	ON
inverter_stale_despite_comms_ok	OFF	ON	ON
door_triggered_alone	OFF	ON	ON
door_triggered_with_bms_comms_lost	OFF	ON	ON
relay_stuck_open	OFF	ON	ON
relay_stuck_closed	OFF	ON	ON
grid_absent	ON	OFF	OFF
grid_out_of_band	OFF	ON	ON
multiple_simultaneous_faults	OFF	ON	ON

The native run checks the supplied snapshots and the `write_relay(false)` shutdown request. Wokwi output checks are reported in Section D.

Appendix B Component list and procurement links

Manufacturer part numbers and buying links are listed below. Check current stock before ordering. Coil equivalents and simulation monitors are not hardware BOM items.

Ref / quantity	Manufacturer and part	Specification and distributor
Q1 / 1	Infineon IRL540NPBF	100 V N-MOSFET, TO-220; 1=G, 2=D, 3=S; DigiKey Product page
TVS1 / 1	Littelfuse SMBJ18A	Unidirectional, 18 V standoff; 29.2 V specified clamp; DigiKey Product page
D1 / 1	Vishay UF4004-E3/54	1 A, 400 V fast rectifier, axial; DigiKey Product page
DZ1 / 1	Vishay BZX55C12-TAP	12 V gate zener, DO-35; DigiKey Product page
D2 / 1	Vishay 1N4148-TAP	Reverse LED protection diode, axial; DigiKey Product page
U1 / 1	Vishay VO617A-4	CTR group 4, DIP-4 optocoupler; Mouser Product page
F1 / 1	Littelfuse 0287001.PXCN	1 A, 32 V DC ATOF; 1 kA interrupt rating; DigiKey Product page
F1 holder / 1	Littelfuse 0FHA0001ZXJ	ATO inline holder, 32 V / 20 A; DigiKey Product page
RG, ROUT / 2	Vishay MRS25000C1001FCT00	1 kohm, 1%, 0.6 W axial; DigiKey Product page
RPU / 1	Vishay MRS25000C1002FCT00	10 kohm, 1%, 0.6 W axial; DigiKey Product page
RPD / 1	Vishay MRS25000C1003FCT00	100 kohm, 1%, 0.6 W axial; DigiKey Product page
RLED / 1	Vishay MRS25000C1801FCT00	1.8 kohm, 1%, 0.6 W axial; DigiKey Product page
CF / 1	KEMET C315C103J1G5TA	10 nF, 5%, 100 V C0G radial; DigiKey Product page
CDEC / 1	KEMET C315C104K5R5TA	100 nF, 10%, 50 V X7R radial; DigiKey Product page

Appendix B Connector parts

Ref / quantity	Manufacturer and part	Specification and distributor
J1 header / 1	JST B5P-VH(LF)(SN)	5-way VH, 3.96 mm; distributor base name B5P-VH; DigiKey Product page
J2 header / 1	JST B4P-VH(LF)(SN)	4-way VH, 3.96 mm; base name B4P-VH; DigiKey Product page
J3 header / 1	JST B3B-XH-A(LF)(SN)	3-way XH, 2.5 mm; Mouser Product page
J1 housing / 1	JST VHR-5N	5-way mating housing; DigiKey Product page
J2 housing / 1	JST VHR-4N	4-way mating housing; DigiKey Product page
J3 housing / 1	JST XHP-3	3-way mating housing; DigiKey Product page
J1/J2 contacts / 8	JST SVH-21T-P1.1	18–22 AWG; fitted 4 + 4; listed CT MOQ 100; DigiKey Product page
J3 contacts / 3	JST SXH-001T-P0.6	22–28 AWG; RS 123-5947 tape MOQ 250; RS Product page

Fit eight contacts for J1/J2 and three for J3. Ordering pack sizes may differ. These connectors suit an enclosed LV prototype. Sealing and vibration suitability depend on the installation. Wire length, PCB, enclosure and tools depend on the team's layout.

Appendix C File organisation and reproduction

Location	Contents and origin
reports/	Combined Word and PDF submission.
src/logic.c; SUMMARY.md	Current implementation and existing summary; copied unchanged.
include/; src/main.c; src/mock_hw.c; test/	Original team harness and predefined scenarios, preserved.
CMakeLists.txt; pico_sdk_import.cmake	Original build configuration; use NATIVE_BUILD=ON for a desktop run.
diagram.json; wokwi.toml	Original Pico SDK configuration. The tested web copy is in wokwi_web/; screenshots are in evidence/D/.
simulations/B_precharge/	Three numerical-node RC precharge decks.
simulations/C_driver/	V5 driver/feedback and fault-case decks, including monitor/supplement cases.
evidence/	Proposed schematics, original cropped simulation screenshots and fresh native log.
components/	Procurement list; historical stock snapshot retained separately.
SUBMISSION_README.md; manifest.sha256	How to reproduce results and check the file list.

In Multisim, load a .cir file with source and its actual full path. Run the matching transient command, then plot the quantities below. The source command loads a netlist for analysis. It does not display a graphical schematic. My C schematics are supplied separately in evidence/C/.

B precharge command example

```
source <full-path-to-B1_V2_Normal.cir>
tran 100u 1.2 0 100u uic
plot v(1) v(2)
plot v(3)
```

Repeat with B1_V2_Open.cir or B1_V2_Short.cir. Read v(3) at 1 V/A. Replace the path placeholder with your actual file path. Do not type the angle brackets.

C driver command example

```
source <full-path-to-C_V5_Normal.cir>
tran 10u 100m 0 1u
plot v(7) v(6)
plot i(lcoil)
plot v(5)
plot v(25) v(26)
```

Load and simulate each scenario before plotting. For the coil-voltage supplement, load C_V5_Normal_Monitors.cir and plot v(27). If a vector is missing, check which netlist is loaded and whether simulation has run. Do not reuse a plot from another circuit.

I checked the buzzer command and GPIO level. The button is sampled at each scenario boundary. These tests assess software decisions and simulated outputs.

For Wokwi, I combined the existing C sources into wokwi_web/main.c. I added sketch.ino for Arduino serial output and start-up. The original fault logic and scenario values are retained. The adapter uses Serial in place of the unsupported stdio_init_all call. It also adds a short start-up delay. AI helped assemble the simulation copy and write the adapter. Original src/logic.c and SUMMARY.md are unchanged.

Wokwi project: <https://wokwi.com/projects/477397690398701569>. A local copy of the web files and the two screenshots is included in the submission folder.

Wokwi reproduction

Open a Raspberry Pi Pico Arduino project in Wokwi. Use the sketch.ino and diagram.json files from wokwi_web/. Add main.c from the same folder. Save and start simulation. Watch one full scenario cycle. Hold DOOR/TAMPER during a nominal_operation evaluation to check shutdown. Release it and wait for a later nominal_operation evaluation to check recovery. The original CMake project remains the native/Pico SDK submission source.